

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifica-
8 tion demonstrating how cognitive architectures can emerge as attrac-
9 tor states from established physical primitives, without introducing
10 new fundamental physics. Starting from the finiteness of a localized
11 system’s phase space, we construct a framework for predictive process-
12 ing; the spacetime symmetries are the setting rather than a premise,
13 and no theorem consumes them. Spontaneous symmetry breaking gener-
14 ates necessary topological boundaries, and finite phase space sub-
15 jects these boundaries to Landauer’s thermodynamic erasure. Agency

16 arises when systems evolve along paths of least action, minimizing en-
 17 tropy production via internal structural resonance with external per-
 18 turbations. Scaling this dynamic to macroscopic levels aligns with
 19 emerging paradigms in neuroscience—specifically classical electrody-
 20 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
 21 tion, culminating in a topological fixed point: a unified “Self.” We
 22 state the claim narrowly, because the subject matter invites a wider
 23 reading than the work supports: no new physics is derived here. Every
 24 physical result the chain invokes is established—Picard–Lindelöf, Kib-
 25 ble’s classification of topological defects, Landauer’s erasure bound,
 26 Gibbs’ inequality, Still’s bound on dissipation by nonpredictive in-
 27 formation, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s fixed
 28 point. What is new is that they are stated in one formal language,
 29 their consistency mechanically checked, and the points where the chain
 30 relies on stipulation rather than derivation located exactly. The chain
 31 itself composes to a single theorem, whose eight unproved steps are
 32 its arguments rather than sentences about it, and which are jointly
 33 satisfiable rather than jointly empty. The development declares no ax-
 34 ioms and every structure carrying physical content is inhabited by a
 35 witness; the synchronization step is proved from a dynamics for initial
 36 data inside an explicit basin and assumed outside it, and the identi-
 37 fication of a global section of the probability sheaf with the unity of
 38 experience is a stipulation, marked as such throughout rather than
 39 absorbed into the derivation. The step from a coarse-grained cou-

pling to the brain’s endogenous electromagnetic field is likewise an empirical commitment and not a deduction; we separate it from the coarse-graining theorem that precedes it, weaken it to a modulatory-but-integrative role, and give four conditions that would falsify it. The framework is nonetheless falsifiable in one place that is its own and no rival’s: recovery from sudden awakening should follow a delayed sigmoid rather than an exponential, because the variable gating it is astrocytic volume on a timescale of seconds to minutes rather than cortical phase realignment on a timescale of milliseconds; and, more sharply, the concentration and the coherence of the phase distribution must collapse onto a single parameter-free curve, the Bessel ratio, throughout that recovery. As a corollary, we show why rigid von Neumann architectures (e.g., GPUs) are thermodynamically disadvantaged or structurally constrained from reaching these specific continuous thermodynamic attractors.

1 Introduction

The pursuit of a physical theory of consciousness typically involves searching for the biological correlates of mind or treating subjective experience as an abstract computational algorithm. This paper asks a more fundamental question: If we assume only basic physical primitives—continuous symmetries (the Poincaré group), geometry, and thermodynamics—is the emergence of a unified, perceiving entity a natural physical attractor?

62 We propose a framework where the structural properties of consciousness
 63 are derived as physical attractor states. Establishing a deductive chain from
 64 basic principles, we demonstrate how continuous symmetries break to form
 65 boundaries, how those boundaries dissipate free energy as non-equilibrium
 66 steady states, and how their evolution along paths of least action results
 67 in physical resonance with their environment. This eventually scales into
 68 the macroscopic electromagnetic unity proposed by emerging field theories
 69 of mind. To avoid critiques of “math theatre,” we delineate between literal
 70 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
 71 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 72 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
 73 spective aligned with Russellian Monism: objective measurements of macro-
 74 scopic electromagnetic fields constitute the third-person “outside,” while the
 75 system’s internal physical deformation to reach structural resonance consti-
 76 tutes the first-person intrinsic experience.

77 **What is new here, and what is not**

78 A framework paper about consciousness invites a particular suspicion, and it
 79 is better answered at the start than at the end. **No new physics is derived**
 80 **here.** Every physical result the chain invokes is established, and most of
 81 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 82 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 83 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound

84 on the heat cost of erasure (Landauer, 1961); Gibbs' inequality for the relative
 85 entropy (Cover & Thomas, 2006); Barbalat's lemma for the asymptotics of a
 86 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
 87 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
 88 Bell and Crooks relating dissipation to nonpredictive information (Still et
 89 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
 90 dissipated work with the relative entropy between forward and time-reversed
 91 ensembles (Kawai et al., 2007); and Banach's fixed-point theorem (Banach,
 92 1922). A physicist reading this paper will recognize all of it and would
 93 establish the point in three paragraphs if we did not. Conceding it costs
 94 nothing.

95 What is new is of a different kind, and it is a smaller claim than a theory
 96 of consciousness usually makes. These results have been stated in one formal
 97 language, their mutual consistency has been mechanically checked, and the
 98 places where the chain relies on stipulation rather than derivation have been
 99 located exactly. The value of a machine-checked chain is not that any link
 100 is novel; it is that the claim being made *about* the chain is itself checkable,
 101 and we state it in the form a reader can check rather than in the form that
 102 asks to be believed. Four things. The development declares **no axioms**:
 103 **#print axioms** on any result reports only **propext**, **Classical.choice** and
 104 **Quot.sound**. Every physical postulate is a field of a class that each model
 105 discharges for itself rather than a global assertion, and every such class is
 106 inhabited by an explicit witness, so no theorem is vacuous for want of a model.

107 The chain composes to a *single* theorem — `chain`, in `Chain.lean` — which
108 derives the Self from the finiteness of the phase space and takes each unproved
109 step as one of **eight named hypotheses**: four formalization gaps, two
110 modelling assumptions and two physical commitments, each an implication
111 between two links rather than an opaque symbol, so that deleting one fails
112 the build instead of softening a sentence. And the eight hold *simultaneously*
113 on one substrate (`chain_nonvacuous`), which rules out the failure in which
114 a composition proves its conclusion only because its premises cannot all be
115 met at once. The seams — where physics stops and interpretation begins —
116 are thereby fixed in the source rather than negotiable in the prose, and their
117 number is a number rather than an impression.

118 Composing the chain also refuted two of its own arrows, and Figure 1
119 draws the corrected picture rather than repairing it quietly. The capacity
120 bound of the first link has no hypotheses at all, so an implication from it to
121 the existence of a boundary is equivalent to assuming the boundary outright:
122 those two links are independent starting points and not consecutive steps.
123 What finiteness does imply directly is the heat cost, skipping the boundary
124 entirely. Neither of the two arrows that are theorems is one the chain’s
125 usual narration draws. Section 2.1 reports what the check cost: three of five
126 original axioms each proved **False**, from one shared root cause, and none of
127 the three had been caught by reading.

128 The framework is not thereby unfalsifiable, and one of its predictions is
129 its own rather than a rival’s. Section 11 argues that recovery from sudden

130 awakening should follow a *delayed sigmoid* rather than an exponential, be-
 131 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
 132 clearance, on a timescale of seconds to minutes, while the cortical rhythms
 133 whose realignment is usually taken to constitute the recovery move in mil-
 134 liseconds. That is a claim about functional form; it is fittable against data
 135 of a kind already collected; and a purely electrical account of the same re-
 136 covery predicts a different form. Because a delayed sigmoid is what any
 137 slow saturating gate would give, the section also states the sharper claim the
 138 same mathematics supports: at every instant of the recovery the concentra-
 139 tion and the coherence of the phase distribution must lie on the fixed curve
 140 $r = I_1(a)/I_0(a)$, the same curve for every subject, with no free parameters
 141 and with the threshold read off the ratio a/r rather than fitted.

142 2 Formalization in Lean 4

143 A core contribution of this work is the verification of our deductive chain
 144 using the Lean 4 interactive theorem prover. This approach builds on recent
 145 efforts to formalize consciousness and brain semantics, such as the 'OnlyOne'
 146 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,
 147 2026). It is necessary to explicitly distinguish which claims have been math-
 148 ematically derived as theorems within the proof assistant, and which remain
 149 irreducible physical postulates. Figure 1 shows the chain with the status of
 150 every node *and of every arrow* marked, and Table 1 states in seven rows

151 what each link assumes. The complete claim-by-claim map to Lean identi-
 152 fiers, which runs to several pages, is Table S1 in the supplement; the main
 153 text does not depend on reading it.

154 2.1 Soundness of the formalization

155 Formalization is only as trustworthy as the axioms it rests on, and an axiom
 156 that is inconsistent silently makes every theorem downstream of it vacuous
 157 while leaving the build green. We report a case of exactly this, found by
 158 auditing our own development, because the failure mode is easy to reproduce
 159 and we have not seen it discussed in the applied-formalization literature.

160 The failure has one shape, and naming it is more useful than cataloguing
 161 its instances: *an axiom that constrains a symbol it does not itself bind*. Such
 162 a declaration is not a postulate about one system. It is a constraint on every
 163 model of the class the symbol belongs to, including models built to violate
 164 it, and a class that has such a model is thereby inconsistent. Three concrete
 165 cases, each of which proves `False` on its own:

- 166 • `spontaneous_symmetry_breaking_pointwise_min` took the potential-
 167 energy functional as an *implicit* argument occurring only in its hypoth-
 168 esis `PotentialEnergy phi = V v0`. Instantiating that functional with
 169 the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields
 170 the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$
 171 this proves $1 = 0$.

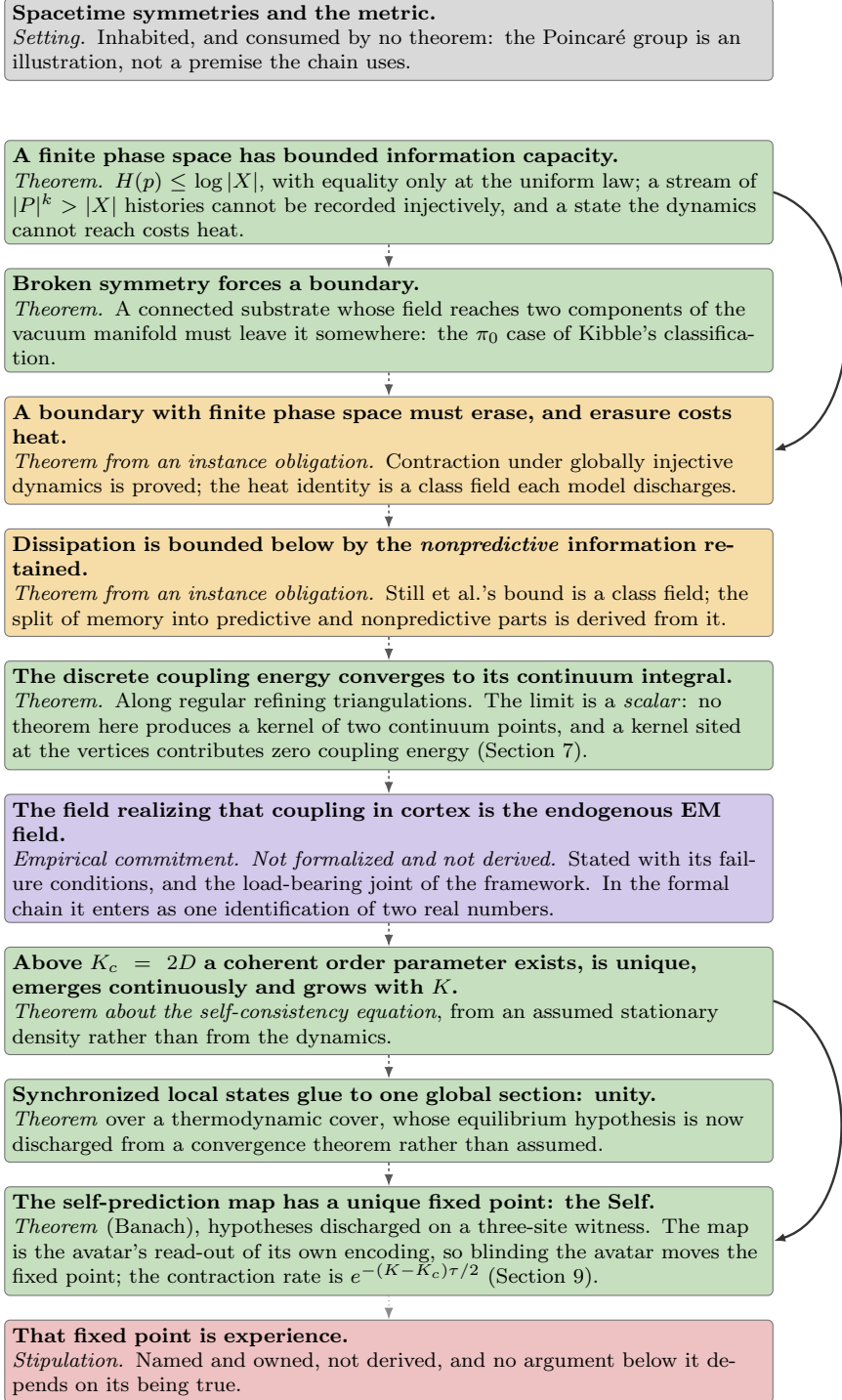


Figure 1: The deductive chain, with⁹the status of each node and of each arrow. *Nodes.* ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ setting, consumed by no theorem. *Arrows.* A **solid** arrow is a Lean theorem. A **dashed** arrow is a named hypothesis of chain (Chain.lean): an implication between the two node statements which the development does not prove and which is therefore an argument the theorem

Claim	Status	What is assumed
A finite phase space has bounded information capacity	Theorem	Finiteness of the state space, and nothing else — no symmetry group, no metric. $H \leq \log X $, with equality only at the uniform law. On an infinite state space the step from “unreachable state” to “erasure” fails, and the failure is exhibited rather than assumed away.
Broken symmetry forces a boundary	Theorem	Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.
Erasures costs heat	Theorem + class field	Global system–bath dynamics injective (Liouville). The heat identity <code>heat.eq</code> is discharged per instance, not asserted for all models.
Dissipation is bounded below by nonpredictive information	Theorem + class field	Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on.
The discrete coupling energy converges to its continuum integral; the field carrying that coupling is the cortical EM field	Theorem; then empirical	Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The limit is a scalar: no theorem produces a kernel of two continuum points, and a kernel sited at the triangulation’s vertices contributes zero coupling energy on an atomless substrate. The physical identification is a commitment with stated failure conditions (Section 7).
Above $K_c = 2D$ a unique coherent state exists, and synchronized patches glue into a unique global section	Theorem 10	Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated. Gluing needs a cover at the potential minimum whose synchronized patches agree on overlaps; the first is derived from a convergent trajectory, the second is an instance obligation throughout.
The coherent order parameter carries a unique reflexive fixed point:	Theorem	The self-prediction map is the avatar’s read-out of its own encoding, and its Lipschitz rate

- 172 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
173 a bath-entropy difference *for every instance* of the class. An instance
174 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
- 175 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P||Q)/\Delta t$ for *all* P and Q , while σ
176 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
177 point mass against a uniform distribution refutes it.

178 Two further declarations of the same shape are worth recording be-
179 cause their refutation is different in kind. `phase_invariant_periodic` and
180 `sync_to_section_eq`, as global assertions pinning free fields of `LocalSectionSynchronization`,
181 are refutable as soon as the probability presheaf admits two distinct global
182 sections — so the refutation needs no arithmetic at all, only that the sheaf is
183 not trivial. Neither is a postulate in the development: the periodicity condi-
184 tion is an obligation of the constructor that builds a cover from an invariant
185 measure, and `sync_to_section_eq` is a theorem about the global section the
186 sheaf condition produces (Derivation 5).

187 The rule that prevents it is uniform, and we state it as the rule we follow:
188 *a physical postulate that mentions a class field must be a field of that class,*
189 *never a standalone axiom quantified over all instances.* A postulate is then
190 an obligation each model discharges, not a global claim about every model.
191 Under that rule the development declares no axioms at all. Two of the
192 statements above are not postulates in the first place — pointwise vacuum
193 minimization is proved from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs'

inequality is Mathlib’s — and the genuinely physical ones, Landauer’s heat identity and the two information bounds, are instance obligations.

This makes non-vacuity the live question, since an uninhabitable class makes its theorems vacuous just as surely as an inconsistent axiom did. `Examples.lean` therefore constructs witnesses: a one-bit erasure system with a genuine bath that discharges `heat_eq` and reproduces $\Delta Q = k_B T \ln 2$; a perfectly resonant system discharging `kl_bound`; and a scalar field on a one-point spacetime discharging `ActionPrinciples`. Derivation 5’s gluing structures are the most demanding case and are witnessed too: `Examples.lean` builds a `ThermodynamicCover` over a three-site substrate carrying the discrete topology and its Borel σ -algebra, covered by two *overlapping* open patches whose intersection is a single shared site. The local sections are germ families of a phase-indexed finite measure $\mu_\theta = (1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the family is provably non-constant in θ (`phaseMeasure_not_const`), so the periodicity obligation of the constructor `LocalSectionSynchronization.ofInvariantMeasure` is discharged by an argument about `cos` rather than satisfied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential` rather than assumed. `global_section_from_thermodynamics` then fires on this instance and glues the two patch-local sections into a unique global one. Two further covers answer the follow-up question of how non-degenerate such a witness can be made. `Examples.lean` §13 has a negative half worth recording: the phase field can be made non-constant as a function, but not as a

217 physical state, because `thermodynamic_equilibrium` forces every instance
 218 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 219 round, from two patches with independently chosen mass profiles that meet
 220 only on the site they share, and the section the gluing returns is neither of
 221 them. That is the same kind of result as `contracting_implies_const` in
 222 Derivation 6 — a proof that the witness is as strong as the class permits,
 223 rather than a witness that could be improved by trying harder.

224 What the witnesses establish is that the classes are inhabitable, so the
 225 theorems about them are not vacuous. What they do not establish is that
 226 a cortex is such an instance. In the cover above the phase field is already
 227 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 228 the equilibrium obligation dischargeable; getting a physical system into that
 229 configuration is the content of the informal dynamical argument, not of the
 230 Lean development.

231 The same audit turned up a defect of a different kind, worth record-
 232 ing because it is not an axiom problem and would not have been caught
 233 by `#print axioms.mesh_refinement_convergence` — the claim that dis-
 234 crete edge weights converge to the continuous energy as the mesh is refined
 235 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
 236 not merely unproved but false. Nothing required the edge regions to cover
 237 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
 238 all empty satisfied the mesh hypothesis while contributing nothing to the
 239 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it

240 quantified over a triangulation and then resolved the vertex type by instance
 241 search rather than through the bound variable, so the variable was never
 242 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
 243 gence is now stated along a *sequence* of triangulations of a named region
 244 S , each required to be regular — edge regions measurable, without self-
 245 loops, disjoint across distinct edges, covering S , and anchored to the sim-
 246 plicial data in the sense that a vertex lies in the closure of every region
 247 of an edge at it. Closure rather than membership is forced: cells are dis-
 248 joint, so a shared vertex belongs to at most one of them. The manuscript’s
 249 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
 250 and is therefore not symmetric, is shown to equal exactly the midpoint rule
 251 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
 252 tion, reading the double sum as a Riemann sum would be a category error.
 253 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 254 the integrand, and convergence follows for uniformly continuous integrands
 255 as the mesh shrinks. The regularity conditions pull against each other, so
 256 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into
 257 half-open cells and checks that the approximation genuinely moves — $\frac{1}{2}$ at
 258 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 259 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 260 the accompanying simulation measures; that would need a midpoint error
 261 term with a second derivative, which we have not developed.

262 A third defect was one of bookkeeping across files, and it is the kind that

survives any number of green builds because each half is individually correct. The development carried *two* Kuramoto potentials. Lyapunov descent along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) = -\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum — phase-locked in both directions — and everything downstream of it, including Derivation 5’s gluing, concerned the coupling term alone. Nothing chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below` now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below — walk θ out along ω and the bounded cosine sum cannot compensate — so it has no minimum for phase-locking to attain, and the sentence “the phase-locked state minimizes the Lyapunov potential” was, read literally, false of the functional the descent theorem was about. `Phase4_RotatingFrame.lean` closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-frequency system — which is exactly the system Derivation 5 already worked with — the two potentials coincide there, and the descent result transports onto the coupling potential whose minimizers are the phase-locked states. Phase differences, phase-locking and the magnitude of the order parameter are all proved invariant under the change of frame, so nothing observable is lost in the reduction. `Examples.lean` §7 runs the chain end to end on two oscillators with a common frequency, discharging every hypothesis. The restriction is real and we state it plainly: the reduction is exact only for identical frequencies. A genuine spread of natural frequencies leaves residual

286 detunings in the reduced system, phase-locking then requires coupling above
 287 a critical K_c , and that K_c — Kuramoto’s threshold for a spread of frequen-
 288 cies, not the noise threshold $2D$ of Section 10 — is nowhere in our Lean
 289 development. It is studied numerically, in Figure 5, as Table S1 records.

290 A fourth defect was dimensional, and it concerned the one row of Table S1
 291 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 292 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 293 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 294 against an unnormalized measure carries the substrate’s mass twice over, so
 295 a constant kernel K on a substrate of total mass m contributed Km^2 . The
 296 comparison therefore depended on how large the substrate was rather than
 297 on how strongly it was coupled, and any kernel whatever could be pushed
 298 above threshold by inflating m . `Examples.lean` §9 exhibits the failure con-
 299 cretely: on a two-site substrate carrying counting measure, a system coupled
 300 at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two *below* threshold
 301 — has an unnormalized double integral of $2 > 1$, and would satisfy any pred-
 302 icate stated with that integral. The fix is to require the substrate’s measure
 303 to be a probability measure, which is the normalization in which the mean-
 304 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 305 then gives that a constant kernel has mean-field strength exactly K , and
 306 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 307 We flag what this does and does not do. It makes the predicate say something
 308 whose truth depends only on the physics, and §9 witnesses it on both sides of

309 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 310 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 311 at $K = 2D$, which is a separate question from whether the predicate is di-
 312 mensionally sound. That bifurcation is proved below, in both directions,
 313 from an assumed stationary density; deriving the density itself needs the
 314 Fokker–Planck operator and is not done.

315 We draw no grand moral from this beyond the obvious one. The value
 316 of a proof assistant here was not that it certified our argument, but that it
 317 made a specific, checkable claim about what had been assumed — and that
 318 claim turned out to be false in a way no amount of reading the prose would
 319 have revealed.

320 **3 Axiom 1: Symmetries as Setting, and the** 321 **Capacity of a Finite Phase Space**

322 Our starting premise relies on fundamental geometry. In our Lean formaliza-
 323 tion (`Phase1Primitives.lean`), continuous spacetime symmetries (e.g., the
 324 Poincaré group) act upon field states, and the invariance of the system’s total
 325 energy under that action is recorded by the class `SymmetryInvariantAction`.
 326 Physically, it is such an invariance that Noether’s theorem converts into the
 327 conservation of energy and momentum. The scope of the *formal* claim is nar-
 328 rower and we state it plainly: the invariance is a class field, i.e. an assumption
 329 a model must discharge, and no conserved quantity is ever constructed from

330 it. There is no Noether theorem in our Lean, and nothing downstream of this
 331 section depends on one. The class is now inhabited — `Examples.lean` §11
 332 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
 333 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
 334 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
 335 changed by the symmetry, so an energy-minimising field and its image have
 336 equal energy and differ. That is symmetry breaking exhibited rather than
 337 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
 338 no one-parameter family to differentiate along in the first place. Simultane-
 339 ously, discrete permutation symmetries dictate the statistical mechanics of
 340 any physical system, defining a strictly finite volume of phase space.¹

341 Because phase space is finite, any localized physical system possesses a
 342 mathematical limit on the amount of information (distinguishable states) it
 343 can embody at any given moment. That sentence is the section’s contribution
 344 to the chain, and three theorems carry it, in `Phase1_PhaseSpaceCapacity.lean`,
 345 whose only hypothesis about the system is that its state space is finite: no
 346 symmetry group, no metric, no action functional appears anywhere in them.

¹While the Poincaré group and discrete permutations serve as accessible physical ex-
 amples, the underlying Lean formalization does not hardcode specific physics. It defines
 the system at a more primitive mathematical level, relying entirely on generic Pseudo-
 Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechan-
 ics). What the formalization supplies is the vocabulary in which symmetries, conserved
 quantities and phase-space constraints can be *stated*; it does not derive them. All three
 of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its in-
 variance in §11, the real line with the Euclidean metric in §12 — but the instances are
 witnesses of inhabitability, not of physics: no theorem in the development consumes a
 symmetry group or a metric, and that half of the premise is setting rather than content.
 The other half of the premise is not setting, and the paragraphs below are what it does.

347 The first is the capacity bound itself. A probability distribution p on a
 348 finite state space X satisfies $H(p) = -\sum_x p(x) \log p(x) \leq \log |X|$ (`shannon_`
 349 `entropy_le_log_card`), with equality if and only if p is uniform (`shannon_`
 350 `entropy_eq_log_card_iff`). The “only if” half is the part worth stating: it
 351 says $\log |X|$ is the exact capacity rather than a convenient over-estimate, so
 352 a system holding any distribution other than the flat one over its states holds
 353 strictly less than its size suggests. The proof is Gibbs’ inequality against the
 354 uniform law — $\log x \leq x - 1$ applied at $x = 1/(|X|p(x))$, with the $p(x) = 0$
 355 case carrying the slack that makes the equality condition sharp — and it is
 356 elementary. What was missing here was the statement, not the difficulty.

357 The second says the limit binds. Let the boundary absorb a stream of
 358 perturbations drawn from an alphabet P ; a stream of length k has $|P|^k$
 359 possible histories. Once $|X| < |P|^k$, the map from histories to final states
 360 cannot be injective (`absorb_not_injective`), so two distinct histories leave
 361 the system in the same state, and the entropy of the source strictly exceeds
 362 anything the system can hold (`source_entropy_exceeds_capacity`). The
 363 update map is arbitrary and is allowed to be reversible at every individual
 364 step; the collision is forced by counting alone, which is the sense in which the
 365 premise, and not the model of the dynamics, is doing the work.

366 The third is what Derivation 2 consumes, and it is the one place where
 367 finiteness does something no other hypothesis could. On a finite state space,
 368 a map that fails to reach every state is not injective (`is_erasure_of_not_`
 369 `surjective`) — which is exactly the erasure condition Landauer’s bound re-

370 quires — so any update of a finite phase space leaving some state unreachable
 371 dissipates strictly positive heat (`finite_phase_space_dissipates`). On an
 372 infinite phase space the step simply fails: the successor map on $\mathbb{N}$ misses 0 and
 373 is injective. `Examples.lean` §19 records that failure next to the witnesses, so
 374 the finiteness hypothesis is fenced rather than assumed to be load-bearing,
 375 and runs all three results on a single bit: the fair coin attains $\log 2$, a $3/4$ –
 376 $1/4$ bit falls strictly short, two binary perturbations already outrun a one-bit
 377 memory, and §1’s one-bit thermodynamic system pays heat for a state it
 378 cannot reach.

379 None of this derives the finiteness of phase space. Finiteness is assumed,
 380 as it is in any statistical-mechanical treatment, and the Poincaré group re-
 381 mains an illustration in prose. What has been removed is the gap between
 382 assuming the premise and using it.

383 4 **Derivation 1: Symmetry Breaking and the** 384 **Inevitability of Boundaries**

385 A perfectly symmetric universe is the uniform vacuum; no distinct entities
 386 exist. For any dynamic process to occur, the continuous symmetries of the
 387 background field must undergo spontaneous symmetry breaking.

388 When a field drops to a lower, asymmetric energy state, topology dic-
 389 tates the natural creation of defects (such as domain walls or monopoles).
 390 This constraint necessitates the creation of a physical boundary—a localized

391 topological distinction between an “inside” and an “outside.” While such
392 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
393 entity, it is not sufficient for agency. The boundary is relational, defined by
394 the twisted geometry of the field surrounding it, and must be coupled with
395 non-equilibrium thermodynamics to exhibit active behaviors.

396 That first sentence is a claim about topology, and it needs a different
397 argument from the energy claim beside it. The energy half — an energy-
398 minimizing field sits in the vacuum manifold almost everywhere (`pointwise_`
399 `vacuum_of_global_min`) — together with the observation that a boundary
400 defect *obstructs* extension over a contractible interior, produces no defect:
401 both are statements about a field already given. What produces one is
402 `exists_notMem_of_connectedComponentIn`: if the substrate is connected
403 and a continuous field takes values in two *different connected components*
404 of the vacuum manifold M , then somewhere the field is not in M at all.
405 That excluded region is the wall. This is the π_0 case of Kibble’s classifica-
406 tion of topological defects by the homotopy of the vacuum manifold (Kibble,
407 1976), and the proof is elementary in exactly the way it is elementary in the
408 physics literature: the continuous image of a connected space is connected, a
409 connected subset of M lies inside one component of M , and the field’s range
410 would be such a subset if it never left M . No hypothesis mentions energy, a
411 potential, or a symmetry; connectedness of the substrate and disconnected-
412 ness of the vacuum are the whole of it.

413 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$

414 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are the rest of Kibble’s
 415 classification and Mathlib’s homotopy-group library would support at least
 416 the next one; they are available and not taken, because the argument this
 417 paper runs uses domain walls and nothing else. The witness is the double
 418 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be
 419 exactly $\{-1, 1\}$ (`wellVacuum_eq`) rather than merely known to contain both
 420 minima, and its disconnection is proved in the form the theorem consumes:
 421 no preconnected subset of it contains both, since a preconnected subset of $\mathbb{R}$
 422 is an interval and an interval spanning -1 and 1 contains 0 , which is the top
 423 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field
 424 on *any* connected substrate that reaches both minima, with no formula for
 425 the field anywhere in the statement. That the class is non-empty is checked
 426 separately, because a theorem quantified over all fields is worthless if none
 427 satisfies its hypotheses: the clipped identity on $\mathbb{R}$ does, and the wall it is
 428 forced to have sits at the origin (`kink_zero_notMem`).

429 **5 Derivation 2: The Thermodynamic Con-** 430 **straint**

431 This newly formed topological boundary is subjected to gauge perturbations
 432 from the “outside” environment. Because the internal phase space is finite
 433 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
 434 and must overwrite old internal states.

435 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
436 formation irreversibly decreases internal entropy, mandating that the sys-
437 tem exhaust heat into the environment. Philosophers of physics (such as
438 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
439 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure
440 (a many-to-one mapping in abstract computational states) with *thermody-*
441 *namic* erasure (a many-to-one mapping in physical phase space). They argue
442 that logical erasure could conceptually be implemented by an injective phys-
443 ical process without dissipating heat.

444 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)
445 addresses this critique by defining erasure as a non-injective mapping on the
446 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
447 tion of the bipartite universe (System + Bath) as strictly injective, repre-
448 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
449 to abstract logical erasure, but is a mandatory mathematical consequence
450 of any local physical state space contraction under globally reversible dy-
451 namics. If a cognitive system’s internal topology forces a contraction of its
452 physical state space to accommodate structural resonance, heat dissipation
453 becomes a thermodynamic requirement. The finiteness premise enters at ex-
454 actly this point. What a modeller can usually check about a physical update
455 is not that it is many-to-one but that some state has become unreachable;
456 on a finite phase space those are the same condition, and `finite_phase_`
457 `space_dissipates` is the resulting statement. That identification preserves

458 the Norton–Shenker distinction rather than blurring it: an update that keeps
 459 the incoming perturbation on the record is injective and dissipates nothing
 460 (`keepRecord_injective`), while one that refreshes the register holding it is
 461 not (`absorbStep_is_erasure`). Losing *which* history occurred is forced by
 462 counting alone and is free; overwriting the register that held it is what is
 463 charged $k_B T \ln 2$. A stable boundary thus constitutes a dissipative structure
 464 (Prigogine & Stengers, 1984), operating as a non-equilibrium steady state
 465 that consumes free energy to pay the thermodynamic cost of erasure. Contin-
 466 uous thermal fluctuations play a constructive role in establishing boundaries
 467 between states while contributing to the system’s flexible, adaptive response.

468 **6 Derivation 3: Prediction as the Thermody-** 469 **namic Cost of Memory**

470 A dissipative structure is driven by its environment. Write X_t for the state
 471 of the structure and S_t for the environmental signal driving it, and let the
 472 signal evolve by a dynamics of its own, $S_t \rightarrow S_{t+1}$, on which the structure
 473 does not act. Two quantities are then defined at each step: the *memory*
 474 $I(X_t; S_t)$, the mutual information between the structure’s present state and
 475 the signal’s present value, and the *predictive information* $I(X_t; S_{t+1})$, which
 476 is what that same state says about the signal’s next value.

477 **What the entropy-production bound does not give.** There is a nat-
478 ural route from entropy production to predictive processing, and it does
479 not work. Model the environment as a stochastic driving force $P(\text{ext})$ and
480 the structure’s internal transitions as $Q(\text{int})$. Entropy production is then
481 bounded below by $D_{KL}(P||Q)/\Delta t$ — this is Gibbs’ inequality, it is carried
482 in Lean as an instance obligation, and §3 of the supplement states it. Re-
483 arranged it reads $D_{KL}(P||Q) \leq \Delta t \sigma$, so a system descending on σ lowers
484 a *bound* on the divergence, and it is tempting to conclude that the internal
485 transition probabilities come to mirror the environment’s statistical symme-
486 tries. They do not, because a driven system does not relax to $\sigma = 0$. It
487 relaxes to a non-equilibrium steady state at strictly positive entropy produc-
488 tion, and the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number.
489 The inequality is true and does real work — combined with Gibbs it gives
490 $\sigma \geq 0$ — but no limit may be read off it, and a framework that wants predic-
491 tive processing has to get it from somewhere else. It gets it from the bound
492 below.

493 **Where predictive processing comes from: a published bound.** Still,
494 Sivak, Bell and Crooks (Still et al., 2012) prove that for a system driven by
495 an environmental signal the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

496 The bracket is the *nonpredictive* information: what the structure retains
 497 about the signal's present that says nothing about the signal's future. Dis-
 498 sipation is bounded below by it. Memory that fails to predict is not merely
 499 useless to the structure; it is thermodynamic waste, and the waste is quan-
 500 tified in energy.

501 **Three consequences, and they are theorems rather than readings.**

502 The bound is carried in Lean as a class field, `PredictiveDissipation.`
 503 `still_bound (Phase3_PredictiveThermodynamics)`, for the same reason ev-
 504 ery other physical postulate here is: its `axiom` form, quantified over all joint
 505 laws with the dissipated work fixed, is refutable, and **Examples** §18.4 refutes
 506 it. What is *derived* is everything downstream.

507 First, *the bracket is non-negative*, and this is proved rather than assumed.
 508 If the structure does not act on its environment, then $X_t \rightarrow S_t \rightarrow S_{t+1}$
 509 is a Markov chain and the data processing inequality gives $I(X_t; S_{t+1}) \leq$
 510 $I(X_t; S_t)$: a system cannot know more about the signal's future than about its
 511 present (`predictiveInfo_le_mutualInfo`). Mathlib carries the Kullback-
 512 Leibler divergence but no mutual information, so the latter is defined here
 513 as the divergence of the joint law from the product of its marginals, which is
 514 the standard definition and makes the library's data processing lemma apply
 515 directly.

516 Second, *the second law follows* instead of being assumed. Dissipation is
 517 bounded below by a quantity now known to be non-negative, so $W_{\text{diss}} \geq 0$

518 (`dissipatedWork_nonneg`). This mirrors the role Gibbs' inequality plays
 519 for the entropy-production bound, with one difference that matters: Gibbs'
 520 inequality compares a system to an arbitrary reference distribution, whereas
 521 the data processing inequality compares the system to its own future, which
 522 is a quantity selection can act on.

523 Third, and this is the point of the replacement, *the inference now runs*
 524 *in the valid direction*. The quantity of interest sits on the small side of
 525 the inequality: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$. No limit is required.
 526 Any bound on dissipation is immediately a bound on wasted memory, so a
 527 structure held to an energy budget is thereby held to a prediction standard:
 528 this is `nonpredictive_le_dissipation`. In the idealised quasi-static case
 529 $W_{\text{diss}} = 0$, the theorem `predictive_eq_of_no_dissipation` gives that every
 530 retained bit is predictive. This is a weaker statement than the one it replaces,
 531 and it is the one that is true.

532 **The hypothesis is load-bearing, and we exhibit what breaks with-**
 533 **out it.** The data processing step requires that the structure not write to
 534 its environment. In the formalization this is not an assumption stated beside
 535 the theorem but the shape of the evolution itself: the signal's step is applied
 536 to the pair as the parallel kernel $\text{id} \parallel \kappa$, which acts as the identity on the
 537 system coordinate and through κ on the signal coordinate, so no channel
 538 from X_t to S_{t+1} exists. **Examples** §18.3 shows the restriction is not decora-
 539 tion: a Markov kernel on the *pair* which copies the system's state into the

540 signal carries an independent two-bit law to a perfectly correlated one, rais-
 541 ing mutual information from zero to a positive value. An agent that steers
 542 its environment can indeed know more about that environment’s future than
 543 about its present, and Still’s bound is not about such an agent.

544 **Both regimes are exhibited, not just described.** Examples §18 builds
 545 two systems over one correlated two-bit joint law, differing only in the signal’s
 546 dynamics. When the signal is frozen, every remembered bit is a predictive
 547 bit, the nonpredictive information is zero, and the instance is permitted to
 548 dissipate nothing — with the memory itself proved strictly positive, so the
 549 conclusion is about a system that does remember something. When the signal
 550 is redrawn from its stationary law at each step, the predictive information is
 551 exactly zero, the *entire* memory is nonpredictive, and the instance is *forced*
 552 to dissipate $k_B T$ times its whole mutual information. Establishing that the
 553 mutual information of the witness is neither zero nor infinite is where the
 554 work of that section lies.

555 **An equality, where the entropy-production route had only a bound.**
 556 Kawai, Parrondo and Van den Broeck (Kawai et al., 2007) show that the
 557 average dissipated work is exactly $k_B T$ times the relative entropy between the
 558 forward and time-reversed phase-space densities. Carried here as a predicate
 559 on an instance rather than as a further postulate, it replaces the abstract
 560 dissipation field with a quantity an experiment can estimate — the statistical
 561 distinguishability of a recording of the system from the same recording run

562 backwards — and yields the statement that the irreversibility of the dynamics
563 is at least the amount of memory that fails to earn its keep (`nonpredictive_`
564 `le_arrow`).

565 **What this does not establish.** It does not deliver the Free Energy Prin-
566 ciple. Still’s bound says that prediction is thermodynamically favoured; it
567 says nothing about hierarchical generative models, variational free energy,
568 or Bayesian inference, and we do not claim to have derived any of them.
569 The Free Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy,
570 2013; Clark, 2016; Spisák & Friston, 2026) and the General Resonance The-
571 ory of Consciousness (Hunt & Schooler, 2019) are consonant with what is
572 proved here, which is a weaker relation than derivation and the only one
573 we assert. Nor does it solve the symbol grounding problem (Harnad, 1990).
574 Maximal predictive information is compatible with the structure storing a
575 lossy, compressed or otherwise unrecognisable function of the signal; a state
576 that predicts is not thereby a state that means. The narrower claim is the
577 one worth making: whatever internal states a physical structure holds, those
578 that do not help predict its environment are paid for in heat, so the pressure
579 toward states indexed to the environment is energetic rather than semantic.
580 We continue to call the phenomenon *structural resonance*, and there is no
581 teleological drive to “survive” in it: predicting the environment is simply the
582 cheaper of the two ways to hold a memory.

583 7 Macroscopic Scaling: A Theorem and a 584 Commitment

585 This is the chain’s fourth step, and it is deliberately not numbered as a
586 derivation. Two steps of different kinds meet here and are easy to run to-
587 gether: a coarse-graining argument, which is a piece of mathematics, and
588 the identification of the coarse-grained medium with the brain’s endogenous
589 electromagnetic field, which is an empirical claim about cortex. Presenting
590 the second as though it inherited the licence of the first would make it look
591 like a deduction, and it is not one. Kept apart, the empirical claim is the
592 framework’s central commitment and the place where it is most easily wrong
593 — which is where a claim of this kind belongs.

594 The formal half: coarse-graining is a theorem, and a 595 narrow one

596 A network of discretely coupled units has a coupling energy that is a sym-
597 metric double sum over pairs. `Phase2.MeshConvergence.lean` proves that
598 this sum converges to a continuous integral: along a sequence of regular
599 triangulations of a region S whose edge regions shrink to points, the dis-
600 crete energy $\frac{1}{2} \sum_u \sum_v \mu(R_{uv}) f(x_u)$ converges to $\int_S f d\mu$ (`mesh_refinement_`
601 `convergence`), witnessed on the uniform partition of $[0, 1)$ in `Examples §6`.
602 The identification that makes this a coarse-graining rather than an arbitrary

603 Riemann sum is `sum_sum_mul_of_symm`: because the summand is symmetric,
604 the double sum is exactly the midpoint rule over unordered edges, so the
605 discrete energy is a quadrature of the continuous one rather than a different
606 quantity that happens to converge to it.

607 What this establishes is worth stating exactly, because it is less than the
608 phrase “coarse-graining, governed by the Renormalization Group” suggests.
609 It is a statement about an *energy functional*, not about a dynamics: nothing
610 here shows that trajectories of the discrete Kuramoto system converge
611 to solutions of a field equation, and the earlier appeal to the Renormalization
612 Group was decorative — no renormalization group flow is constructed
613 anywhere in this development. The limit object is a scalar density on the sub-
614 strate, in the intended reading the magnitude of the stress-energy tensor, and
615 it is not an electromagnetic field. The continuous object the later sections
616 use, `ContinuousNeuralField`, is *posited* as a structure with a coupling kernel
617 K ; no theorem produces it from the discrete system. The gap between the
618 finite Kuramoto model and its continuum limit is the propagation-of-chaos
619 problem, which we have not addressed and record as open.

620 **The empirical half: which physical field realizes the ker-** 621 **nel**

622 The formal half says that a large network of symmetrically coupled units
623 admits a continuum description with a symmetric kernel. It says nothing

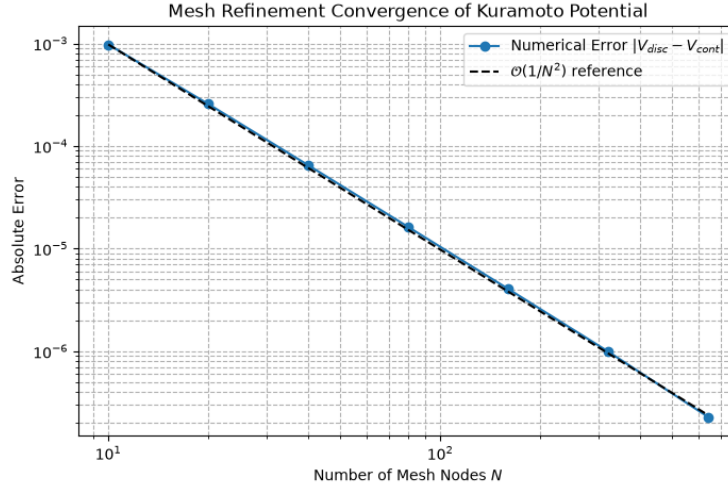


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of N units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field $\theta(x) = \sin x$, is compared against the continuous integral it is proved to converge to. The error falls as $\mathcal{O}(N^{-2})$, the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones.

624 about which physical field in cortex that kernel describes. Our answer is an
 625 empirical commitment, and we state it as one:

626 *In the mammalian cortex, the field realizing the coarse-grained*
 627 *coupling kernel is the endogenous electromagnetic field.*

628 This is a claim about biology, not a consequence of anything above it.
 629 It is what makes the framework falsifiable and it is what we expect to be
 630 attacked.

631 **What supports it.** Neurons do not only signal through synapses: trans-
632 membrane currents generate extracellular fields that shift the membrane po-
633 tentials of neighbouring cells, a mechanism known as ephaptic coupling. En-
634 dogenous fields of 1–5 mV/mm have been shown *in vitro* to entrain spike
635 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015); realistic
636 head models put mesoscopic ephaptic effects within the range where they
637 can bias network dynamics (Ruffini & Santarnecchi, 2020); cable-theoretic
638 treatments show field coupling is sufficient for coincidence detection in the
639 auditory brainstem (Goldwyn & Rinzel, 2016); and stable electric fields have
640 been argued to carry information that survives representational drift in the
641 underlying spiking population (Pinotsis & Miller, 2022, 2023a,b; Cunha &
642 Lima, 2024). The field is spatially continuous and superposes contributions
643 from large populations, which is the property the framework needs: it is a
644 medium in which spatially separated activity is combined without a further
645 mechanism for combining it. This is the integrative role the CEMI field
646 theory assigns it (McFadden, 2002, 2020; Pockett, 2012).

647 **What the evidence supports, and what it does not.** The endoge-
648 nous field’s effect on any individual neuron is *modulatory*. Anastassiou and
649 Koch report that endogenous fields of 1–5 mV/mm shift spike timing by one
650 to three milliseconds (Anastassiou & Koch, 2015), an effect that is real, re-
651 producible, and subthreshold: it biases when a cell fires; it does not decide
652 whether it fires. Nothing in the work cited above supports calling the field

653 the *primary* dissipative structure of the system, and we do not claim it. The
654 claim is that the field is *modulatory in its effect on individual neurons and*
655 *integrative in its spatial reach* — weak per synapse, and the only variable in
656 the system that is defined everywhere at once. What the framework needs
657 is the reach, not the strength, and the two are easy to conflate.

658 **Why reach is the property that matters.** A theory of consciousness
659 needs an account of why many local processes make one experience, and most
660 supply it by stipulation or by a measure. The account here is structural, and
661 it is the framework’s strongest argument. The field is the only variable in the
662 system that is *defined everywhere at once*. Unity is not asserted of it; unity
663 is what a field is. That is what makes Derivation 5’s sheaf-theoretic step
664 available at all — local states glue to one global section because there is one
665 object over which gluing can happen — so the answer to the combination
666 problem is a piece of geometry rather than a postulate. Compare “integration” in Integrated Information Theory, which is a number, or “broadcast” in
667 Global Workspace Theory, which is a metaphor about a workspace. It is also
668 what the identification above is carrying: if the coarse-grained coupling is
669 realized by something that is *not* defined everywhere at once — by synaptic
670 connectivity, say — then this argument is unavailable, and the framework
671 loses the property that distinguishes it from its rivals rather than merely
672 losing a mechanism.

674 **Its critics.** The objection with the longest history is that the fields are
675 too weak to matter, and it has been pressed from within the field-theoretic
676 camp: Pockett, who has herself defended an electromagnetic theory of con-
677 sciousness (Pockett, 2012), catalogues its difficulties (Pockett, 2002) — the
678 measurement problem of relating gross field structure to reported experience,
679 the absence of a constant relationship between the two, and the question of
680 whether the field does causal work or merely reflects it. The quantitative case
681 for scepticism has strengthened since. Vöröslakos and colleagues measured
682 intracranial fields during transcranial stimulation in rats and human cadav-
683 ers and found that the great majority of applied current is shunted through
684 the scalp, so that conventional protocols deliver fields at or below the 1 V/m
685 range in cortex (Vöröslakos et al., 2018). That is a bound on *applied* fields
686 rather than endogenous ones, but it constrains how much causal work a field
687 of that magnitude can be doing, and it is the reason we do not claim the
688 field drives spiking.

689 **What would falsify it.** We state the failure conditions rather than leaving
690 them to a referee.

691 1. *Epiphenomenality.* If the endogenous field can be cancelled or shunted
692 — by a conductive medium that nulls the extracellular potential with-
693 out altering transmembrane currents — and neither behaviour nor re-
694 port changes, the identification fails. This is the decisive experiment
695 and it is, in principle, available.

696 2. *A magnitude gap.* If the spatial integration the framework requires
697 demands field effects an order of magnitude above the 1–5 mV/mm
698 measured in cortex, the commitment fails on arithmetic. The relevant
699 number is not whether ephaptic coupling exists but whether its coupling
700 strength suffices to place the system above the critical coupling K_c of
701 Derivation 7.

702 3. *A timescale mismatch in the wrong direction.* The prediction of Sec-
703 tion 11 turns on the field’s dynamics being fast relative to the astrocytic
704 and clearance variables. If recovery from sleep inertia tracks an expo-
705 nential set by the electrical timescale rather than the delayed sigmoid
706 the slow variable predicts, the framework loses the one prediction that
707 is its own.

708 4. *Dissociation of field coherence from reported unity.* If large-scale field
709 coherence is preserved across states in which experiential unity is lost
710 — or absent in states where it is retained — the identification of the
711 global section with unified experience has no physical carrier.

712 A framework that survives all four is not thereby true. It is merely still
713 standing, which is more than most theories of consciousness arrange to be
714 able to lose.

715 **Geometric frustration, and the regime our theorems ac-**
716 **tually cover**

717 The system’s memory capacity is standardly grounded in *geometric frustra-*
718 *tion*: a highly coupled network cannot bring all its nodes to their lowest en-
719 ergy simultaneously, producing a jagged landscape of near-degenerate states
720 with the capacity to store many patterns. That argument is standard and it
721 is not what this development formalizes. Confronting the discrepancy is the
722 honest course, so we do it here rather than in a footnote.

723 Frustration requires competing couplings — in the Kuramoto setting, a
724 sign structure that no phase assignment can satisfy on every edge at once.
725 The formal setting admits this: `KuramotoSystem` carries $A : V \rightarrow V \rightarrow$
726 $\mathbb{R}$, symmetric and otherwise unconstrained, so signed couplings are repre-
727 sentable. But every theorem in this development that says anything about
728 *where* the dynamics ends requires strictly positive coupling. `phase_locked_`
729 `minimizes_potential` assumes $A_{ij} > 0$; `potential_min_iff_phase_locked`
730 inherits it; the class field `ThermodynamicCover.A_pos` demands it of every
731 instance, and hence of every conclusion of Derivation 5; and the convergence
732 theorem of `Phase4_RotatingFrame` §7 requires the strictly stronger uniform
733 bound $0 < a \leq A_{ij}$. A network with all-positive couplings is an unfrustrated
734 ferromagnet whose potential has one minimum up to a global phase shift. It
735 is the opposite of a jagged landscape, and it has no capacity to store multiple
736 patterns at all. **The system in which memory is grounded and the**

737 **system our theorems describe are not the same system.**

738 The division is not arbitrary, and it is worth recording where it falls. What
739 survives with no assumption on the sign of the coupling is the dynamical half:
740 well-posedness on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`, `unique`), co-
741 variance under passage to a rotating frame, monotone descent of the potential
742 (`dynamic_potential_antitone`), finiteness of the dissipation integral, and —
743 through Barbalat’s lemma — that the motion stops along every trajectory
744 (`velocity_sq_tendsto_zero`). *That* the motion stops is frustration-agnostic.
745 *Where* it stops is not, and every claim this framework makes about the des-
746 tination lives on the positive side of that line. On the continuum side the
747 accounting differs slightly: `phase_locked_minimizes_entropy_of_symm` as-
748 sumes only that the kernel is symmetric and so tolerates signed kernels, but
749 it is conditional on a dynamically phase-locked field existing — which is
750 precisely what frustration may prevent.

751 We therefore scope the claim rather than repeat it. The memory capacity
752 of a frustrated cortical network is not supported by any theorem here, and we
753 do not assert it as part of the chain. What the chain uses is the unfrustrated
754 regime, in which synchronization is reached and the global section exists.
755 Whether a frustrated network can hold many near-degenerate patterns *and*
756 still synchronize enough of itself to admit a global section is an open question
757 about mixed-sign oscillator networks, and it is the most substantive gap we
758 know of in this framework. Recording it as such costs a claim we would have
759 liked to make and is preferable to holding two incompatible pictures at once.

760 **The supporting physiology**

761 The continuous medium is physically supported by astrocytic syncytia—non-
762 neuronal networks joined by gap junctions that provide macroscopic cytoplas-
763 mic continuity for energy and ion flows (Verkhatsky & Nedergaard, 2018).
764 Managing the thermodynamic cost of this state requires sophisticated micro-
765 bioenergetics. Dynamic mitochondrial networks function as deep-level Mi-
766 tochondrial Information Processing Systems (MIPS), transducing biological
767 and environmental information to sustain the coherence of the macroscopic
768 field (Picard & Shirihi, 2022). Discrete synaptic firings, coupled with mito-
769 chondrial micro-mechanics, generate and sustain the field. Because the exter-
770 nal environment is continuous, achieving maximal thermodynamic structural
771 resonance is optimally realized within this continuous macroscopic medium
772 (van Bree et al., 2025).

773 **8 Derivation 5: Global Sections and the Unity** 774 **of Consciousness**

775 The system is now characterized by a continuous macroscopic electrodynamic
776 field generated by millions of coupled micro-components, forced to resonate
777 with the environment. Following Section 7, the coupling regime assumed
778 from here on is the unfrustrated one — all couplings positive — which is
779 what every theorem below requires and what the frustrated network of the

780 memory argument is not.

781 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-
782 tors guarantees that, past a critical coupling threshold, these distributed
783 local fields undergo a phase transition into unified synchronization (Bauer et
784 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
785 phase-locked state minimizes the Lyapunov potential of the system, in the
786 precise sense discussed in Section 2.1: the functional minimized is the cou-
787 pling potential read in the frame rotating with the natural frequency, which
788 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a
789 system of identical oscillators.

790 This phase transition mathematically glues local synchronized states into
791 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
792 The EM field acts as a single, indivisible topological space, providing the
793 *Unity of Consciousness*—a necessary condition for integrated experience that
794 solves the binding problem. While Global Workspace Theory (Baars, 1997;
795 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
796 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
797 our derivation provides a continuous physical mechanism. The contrast is
798 worth drawing carefully, because both theories are addressing the same ex-
799 planandum from different directions.

800 We state the scope of the Lean result precisely, because it is narrower
801 than the prose above. `ThermodynamicCover` demands that the cover’s phase
802 field minimize the reduced Kuramoto potential, and `potential_min_iff_`

803 `phase_locked` characterizes those minimizers exactly: they are the phase-
804 locked configurations, in both directions. Every instance is therefore already
805 phase-locked (`ThermodynamicCover.phase_locked`), and the class's single
806 condition on its local data — that patches at a common phase agree where
807 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
808 sections compatible. `probability_glue_unique`, which is the sheaf property
809 of the probability presheaf and contains no physics at all, then glues that
810 family into exactly one global section.

811 The shape of the class is what makes this an emergence theorem rather
812 than a uniqueness theorem, and the difference is easy to lose. Were `LocalSectionSynchronization`
813 to carry a *global* section as a class field, together with a second field declar-
814 ing every local section to be a restriction of it, the global object would exist
815 before the sheaf condition was ever consulted, and `global_section_from_`
816 `thermodynamics` could do no more than rule out competitors to an object
817 the instance had supplied. It carries neither. Every field of the class men-
818 tions a single patch or a single overlap; the global section is constructed
819 (`ThermodynamicCover.invariantMeasure`); and the statement that the lo-
820 cal sections are restrictions of one common measure is a *theorem* about that
821 construction (`ThermodynamicCover.sync_to_section_eq`), true because the
822 sheaf condition says so, not because an instance was required to name a
823 measure.

824 `Examples.lean §14` is the witness that the difference is not notational.
825 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across

826 the three sites, agreeing only at the site they share, where both put mass 1
 827 — which is what discharges the overlap condition, by computing two num-
 828 bers rather than by construction. The section the theorem returns has pro-
 829 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW.ne_leftW`, `gluedW.ne_`
 830 `rightW`), and neither of them restricts correctly to both patches (`leftW_`
 831 `glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
 832 tion that no single patch had. `Examples.lean` §13, built instead through
 833 the constructor `LocalSectionSynchronization.ofInvariantMeasure`, sits
 834 alongside it for the negative result it carries: its phase field is not the con-
 835 stant function — the second patch a full turn ahead of the first — and by
 836 `ThermodynamicCover.phase_locked` that is as far from constant as any in-
 837 stance can get, since no instance can carry phases differing by anything a
 838 measure is allowed to see.

839 Two hypotheses carry physical content. One of them is an obligation on
 840 every instance: that synchronized patches agree on their overlaps is a mod-
 841 elling assumption, not a consequence of any dynamics we formalize. The
 842 other, `thermodynamic_equilibrium`, asks that the cover already sit at the
 843 potential minimum, and no theorem says an *arbitrary* system’s dynamics
 844 takes it there. “A thermodynamic phase transition into unity” is therefore
 845 doing causal work that no general Lean result supports, though not work
 846 without any support at all: `Examples.lean` §15 exhibits a two-oscillator tra-
 847 jectory that starts a quarter turn out of phase and converges to the global
 848 minimum of the dynamic potential, with the order parameter tending to 1.

849 That trajectory is not an isolated construction: `kuramoto_cauchy_problem`
 850 proves the Kuramoto initial value problem well posed on all of $\mathbb{R}$, so through
 851 every initial state there passes exactly one trajectory, and the exhibited one
 852 is the only solution through its own. Nor is its settling down peculiar to
 853 it: along every trajectory of a zero-frequency system the dynamic potential
 854 converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite and equal to the po-
 855 tential's total drop (`dynamic_potential_tendsto`, `dissipation_integral_`
 856 `tendsto`). The motion also stops: a finite integral does not by itself force
 857 its integrand to zero, but the dissipation rate is Lipschitz in time, so Bar-
 858 balat's lemma applies, and every trajectory satisfies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_`
 859 `sq_tendsto_zero`; Mathlib has no Barbalat, so it is proved here as `tendsto_`
 860 `zero_of_lipschitz_of_integral_le`). That last step — locating where the
 861 motion stops — is taken conditionally, and by a route that avoids the LaSalle
 862 invariance principle, which Mathlib does not carry. LaSalle is the tool for
 863 a system with a decreasing functional and nothing else; this one has more,
 864 and the extra structure is what makes the conclusion reachable without it.
 865 The flow is a gradient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an
 866 identity rather than an inequality; the minimisers of V are characterised ex-
 867 actly; and on the region where every pair of phases lies within a quarter turn
 868 a Polyak–Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is
 869 the potential's excess above its minimum and $a > 0$ is a lower bound for the
 870 coupling (`lojasiewicz_estimate`). Its proof is two extremal indices and the
 871 identity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase

872 is pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 873 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 874 as well, because the invariant region is produced by the Lyapunov function
 875 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 876 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 877 continuity on the connected half-line then prevents any pair from chang-
 878 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to
 879 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each
 880 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 881 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 882 `minimum`: if the phases start within a quarter turn of each other and the initial
 883 excess is below $a/2$, the trajectory converges to a phase-locked configuration
 884 which is a *global* minimiser of the dynamic potential, with order parameter
 885 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
 886 reached rather than assumed. `Examples.lean` §17 fires it on three sites with
 887 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
 888 tion: the trajectory comes from the existence theorem and its limit from the
 889 convergence theorem, so neither is written down.

890 `Phase5.EquilibriumBridge.lean` closes the gap that remained between
 891 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
 892 cover whose phase field is the limit of such a trajectory and returns an in-
 893 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
 894 *vergence*; the cover's coupling matrix is the coupling matrix of the dynamics

895 rather than a separately chosen one, so the system whose equilibrium the
 896 class asserts and the system whose trajectory is run are the same system.
 897 **Examples.lean** §17.1 is the resulting witness: three patches over the three-
 898 site substrate, each carrying its own mass profile on its own open set, above a
 899 phase field that no line of the construction writes down. The trajectory that
 900 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (**trioCover_**
 901 **start_not_locked**), so on this cover the configuration Derivation 5 requires
 902 is arrived at rather than posited. The state the three patches glue to has
 903 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (**trioW_**
 904 **ne_glued**), so one cover now carries both the emergence reading of the gluing
 905 and the derived reading of the equilibrium. What is still assumed on it is the
 906 overlap agreement, and nothing else.

907 The two hypotheses cannot be dropped, and this is the honest scope of
 908 the result. Splay and twisted configurations are equilibria of the same flow,
 909 so “every trajectory reaches the unified state” is false as a general statement,
 910 and the initial-data condition is the standard arc condition under which the
 911 true statement holds. What is claimed is therefore conditional — unity is
 912 reached from configurations already sufficiently coherent — and not that an
 913 arbitrary initial state is carried into unity. What Lean establishes about the
 914 gluing itself is the other half: *given* locally synchronized patches, a global
 915 state exists, is unique, and is produced by the gluing rather than presupposed
 916 by it.

917 Integrated Information Theory begins from phenomenological axioms and

918 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
 919 with the exclusion postulate selecting a single maximum- Φ complex as the
 920 locus of experience. Our sheaf-theoretic global section plays a structurally
 921 analogous role — it too picks out a distinguished unified object — but it
 922 is arrived at from the opposite end: not postulated from the character of
 923 experience, but forced by a thermodynamic phase transition in a physical
 924 field. This buys two things IIT does not have. First, the unity condition
 925 is checkable from field dynamics rather than requiring the computation of
 926 Φ , which is intractable for systems of realistic size and whose definition has
 927 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
 928 out rather than being postulated: gluing either succeeds on a given cover or
 929 it does not, so there is no separate principle needed to rule out overlapping
 930 conscious subsystems. It also inherits IIT’s exposures. Critics have argued
 931 that Φ can be high in systems with no plausible claim to experience, such
 932 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable
 933 to the mirror objection, since a physical system could in principle satisfy
 934 the gluing condition without being cognitively interesting. We take this
 935 seriously rather than dismissing it: our reply is that gluing is necessary but
 936 not sufficient, and that the topological boundary of Derivation 1 and the
 937 reflexive structure of Derivation 6 are additional required conditions. IIT, by
 938 contrast, treats Φ as both necessary and sufficient.

939 Global Workspace Theory faces a complementary difficulty. It specifies
 940 a functional architecture — a limited-capacity workspace broadcasting to

941 specialized modules — without specifying what physical property of broad-
 942 casting makes the broadcast content conscious, leaving the theory open to the
 943 charge that it explains access consciousness while leaving phenomenal con-
 944 sciousness untouched (Block, 1995). Our framework can be read as supplying
 945 the missing physical substrate for the workspace: the continuous macroscopic
 946 field *is* the medium in which integration is realized, and the $K_c = 2D$ thresh-
 947 old gives a physical condition for ignition rather than a functional stipulation
 948 of it. Where GWT is agnostic about implementation, we make a falsifiable
 949 commitment — unity requires a continuous physical medium, which predicts
 950 that a functionally equivalent discrete broadcast architecture would not be
 951 conscious. We regard this as a genuine cost of the position, not a free ad-
 952 vantage: it is the point at which our framework departs most sharply from
 953 computational functionalism and where it is most readily falsified. Contrast
 954 this derivation with panpsychism: consciousness is not assumed as a funda-
 955 mental property of matter. It requires the initial existence of a topological
 956 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
 957 dynamic phase transition into unity that occurs when integrated spatial cou-
 958 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
 959 the foundational topological boundary and the subsequent mathematically
 960 defined phase transition, the system remains a fragmented, non-conscious
 961 collection of micro-states. A unified field of perception, however, is not yet
 962 a “Self.”

963 9 Derivation 6: Reflexive Topology and the 964 Emergence of the Self

965 While a global section provides Unity, a “Self” requires reflexivity. The
966 unified EM field is a dissipative structure that must resonate with external
967 perturbations while managing its own internal phase transitions. Time evo-
968 lution naturally drives the field to minimize thermodynamic friction with its
969 own internal state-changes.

970 This mandates *auto-resonance*: the field must predict itself. The “Self”
971 emerges as a higher-order topological feature when the internal geometry
972 folds to include a low-dimensional avatar of its own computational and causal
973 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)
974 (Levin, 2019). The system is unified, predictive, and explicitly maps its
975 own distinct existence from the environment. This architecture provides the
976 physical foundation for a self-aware consciousness, bounded by the continuous
977 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).

978 **What carries the weight.** `reflexive_topology_implies_self` is the
979 Banach fixed-point theorem with the physical commitments in its hypotheses:
980 the space of global sections is assumed nonempty, metric and complete, and
981 the predictive map is assumed to contract. `Examples.lean` §10 discharges all
982 four on the three-site substrate of Derivation 5, and the metric is constructed
983 rather than posited. The section first pierces the sheafification: on a discrete

984 substrate every point has a smallest open neighbourhood, so the mass a germ
 985 carries at a point is well defined, and `massEquivOn` proves that over *any* open
 986 the sections of the probability sheaf *are* the finite measures on that open,
 987 read as densities against counting measure — the global case being $U = \top$.
 988 Mathlib supplies no inverse to this sheafification — its `TopCat` sheafification
 989 has neither an adjunction nor an `isIso_toSheafify` — so the correspon-
 990 dence had to be constructed. The metric on sections is then the uniform
 991 distance between the densities of the measures they glue to — equivalent to
 992 their total variation on a finite substrate, though not equal to it — and it
 993 is proved complete by exhibiting the limit section of a Cauchy sequence of
 994 densities, not by the degenerate route of showing Cauchy sequences are even-
 995 tually constant. The choice of metric is load-bearing rather than incidental:
 996 under the 0/1 metric on sections every contraction is constant (`contracting_`
 997 `implies_const`), so a witness built there would discharge every hypothesis
 998 of the theorem and say nothing. That theorem is kept in the source, because
 999 which metrics make the fixed point trivial is part of what the fixed point is.

1000 **The map is the avatar, or there is no map.** A reflexive boundary could
 1001 be defined with the predictive map as free data sitting beside the avatar, and
 1002 the reason it is not is a theorem rather than a preference. With the map
 1003 free, the fixed-point theorem projects it out and never touches the avatar, so
 1004 replacing the avatar by a constant leaves a legal boundary with the same pre-
 1005 dictive model and the same Self — a fixed-point theorem that cannot detect

1006 whether the avatar reads the field is Banach's theorem with a consciousness-
 1007 flavoured label. The structure is built the other way round. A reflexive
 1008 boundary carries the avatar's *write* — `auto.resonance`, the encoding of the
 1009 global state into the avatar region — and its *read-out*, the global state that
 1010 encoding predicts; the predictive map is their composite and is not a field.
 1011 The self-model therefore runs through the avatar region or it does not exist.
 1012 Three consequences follow that the free-data shape cannot state. The Self
 1013 lies in the range of the read-out (`self.mem.range.readout`), so it is recon-
 1014 structed from a single section over the avatar region: the low-dimensional
 1015 fold this section has always asserted, as a theorem. Under the resonance
 1016 condition below, the Self satisfies $s = \text{readout}(s|_{\text{avatar}})$ (`self.eq.readout_`
 1017 `restrict`) — the field is what its own localized self-encoding reconstructs.
 1018 And blinding the avatar *moves* the Self: the blinded boundary's map is con-
 1019 stant, its unique fixed point is the state its stipulated contents extend to, and
 1020 the original Self is not a fixed point of it (`not_self_of.constResonance`, wit-
 1021 nessed on the cortex by `cortexBlind.self.ne`). The opposite statement —
 1022 that blinding changes nothing — does not typecheck against this structure,
 1023 because a contraction hypothesis about a boundary's own map says nothing
 1024 about the blinded one's.

1025 **Auto-resonance as a constraint.** That the avatar's state *is* the presheaf
 1026 restriction of the global state is stated as a predicate, `IsRestriction-`
 1027 `Resonance`, rather than added to the structure, so that results say which

1028 of them need it. In the positive direction, if a family of resonant avatars *cov-*
 1029 *ers* the substrate then the global state is determined by their local readings
 1030 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
 1031 recovered from them, by the same sheaf condition that Derivation 5 uses.
 1032 In the negative direction a constant avatar fails the predicate as soon as the
 1033 avatar region separates two states of the field. Both halves are witnessed on
 1034 the three-site substrate, where that separation is proved before anything is
 1035 claimed for it, so the condition is a constraint there and not a formality.

1036 **The contraction rate is K and D , not a numeral.** The remaining stip-
 1037 ulation was the rate. “The predictive map contracts by $1/2$ ” was a fact about
 1038 a map we had defined to average with a baseline, not a claim about a sub-
 1039 strate. The rate is now $\exp(-(K - K_c)\tau/2)$ with $K_c = 2D$ (`resonanceRate`)
 1040 — the linear relaxation of the mean-field order parameter over an interval τ ,
 1041 read off the same threshold Derivation 7 (Section 10) proves the phase tran-
 1042 sition at. Two theorems then split the parameter plane. Above threshold
 1043 the rate is below one and Banach applies, so the Self exists and is unique
 1044 (`self_of_supercritical`). At or below threshold the rate is at least one,
 1045 and `ContractingWith` at that rate is false *for every self-map of every metric*
 1046 *space* (`not_contractingWith_resonanceRate`): the argument is not merely
 1047 unproved below K_c but unavailable. We are careful about what this does and
 1048 does not say. It does not say a Self fails to exist below threshold — only that
 1049 this route to one is closed — and the Lipschitz bound at that rate remains

1050 an instance obligation, discharged on the three-site witness at $K = 3$, $D = 1$,
 1051 $\tau = 2 \ln 2$, where the rate is exactly the one half the read-out achieves. What
 1052 changed is that the obligation is stated in the substrate's own parameters
 1053 and its consequence is conditional on the threshold, rather than on a con-
 1054 stant chosen to make the witness work. The link to Derivation 7 is carried
 1055 by a theorem rather than by a shared numeral, and the distinction is worth
 1056 being exact about. `self_of_supercritical` takes $K > K_c$ as its hypothesis;
 1057 K_c is a definition unfolding to $2D$, so that statement is insensitive to whether
 1058 the bifurcation is proved at all. `Chain.lean` states the link in the form that is
 1059 not. `self_of_coherent_order_parameter` takes the *existence of a coherent*
 1060 *order parameter* as its hypothesis and recovers the threshold inequality from
 1061 it, by `supercritical_of_coherent` — the contrapositive of the subcritical
 1062 uniqueness result of Section 10, since a strictly positive solution of the self-
 1063 consistency equation cannot exist at or below K_c . Derivation 6's fixed point
 1064 therefore rests on Derivation 7's theorem.

1065 **What is still not established.** That any field dynamics produces this
 1066 particular read-out or performs the restriction; and that this three-site sub-
 1067 strate's coupling is $K = 3$ — the parameters are a choice of units for the wit-
 1068 ness, not a measurement, and nothing here derives a coupling constant from
 1069 cortex. We also note a negative result found in the course of this work: the
 1070 Lévy–Prokhorov metric, which the natural weak-convergence reading would
 1071 suggest, does *not* support this argument on a discrete substrate, because the

1072 ε -thickening of a set is the set itself for $\varepsilon < 1$ and the relaxation map is
 1073 consequently not a Lévy–Prokhorov contraction.

1074 **10 Derivation 7: Continuous Stochastic Ther-** 1075 **modynamics and Field Theory**

1076 To move beyond finite graph analogies, we model the macroscopic electro-
 1077 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
 1078 continuous spatial phenomenon, not a discrete network. We extend the
 1079 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
 1080 (`Phase8ContinuousField.lean`).

1081 Subject to thermodynamic noise D , the system undergoes non-equilibrium
 1082 stochastic thermodynamics. Rather than a static Landauer bound, the field
 1083 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (2)$$

1084 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
 1085 dergo gradient descent on this entropy production functional.

1086 That gradient descent is a theorem rather than a modelling assertion,
 1087 for finite substrates. Reading σ as a functional of the coupling kernel at
 1088 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative

1089 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (3)$$

1090 and `sigmaOfKernel.eq.entropy_production_rate` identifies the functional
 1091 being differentiated with `entropy_production_rate` itself whenever the sub-
 1092 strate carries counting measure. `structural_resonance_decreases_entropy_production`
 1093 then concludes that σ is non-increasing along any coupling trajectory relax-
 1094 ing down its own gradient at a positive rate c . Two limitations stay explicit.
 1095 The derivation is for *finite* substrates, and the obstacle to the continuum case
 1096 is *not* differentiation under the integral sign with respect to the kernel, which
 1097 is what the shape of σ suggests at first reading. σ is not a general functional
 1098 that happens to be written as an integral — it is a quadratic functional of an
 1099 *affine* image of the kernel, since the drift is a fixed ω plus a linear operator ap-
 1100 plied to K , and σ is a constant times the squared norm of the drift. A squared
 1101 norm is Fréchet differentiable on any real inner-product space and an affine
 1102 map is its own derivative, so the composite differentiates by the chain rule
 1103 with no limiting argument. `hasFDerivAt.quadratic_of_affine` states that
 1104 in full generality, and `hasFDerivAt.sigmaOfKernel_of_operator` re-derives
 1105 the finite gradient from it, showing that the closed form displayed above is
 1106 a convenience rather than the content. What the continuum case needs is
 1107 therefore one object: the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as
 1108 a bounded operator $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz esti-

1109 mate — the integral kernel is bounded by 1, so on a finite measure space the
 1110 operator norm is at most $\mu(M)^{1/2}$. That object is built. `kernelCLM` is the
 1111 operator, for any kernel factor bounded by 1, with `opNorm_kernelCLM` the
 1112 stated bound; `continuumDriftCLM` is the instance with the Kuramoto factor
 1113 $\sin(\theta_y - \theta_x)$. One step beyond the estimate is needed and is worth naming,
 1114 because the estimate alone does not supply it: the descent theorems ask for
 1115 the derivative as a *gradient vector*, not as an arbitrary bounded functional,
 1116 and producing one requires the adjoint, which exists because L^2 is com-
 1117 plete. `hasFDerivAt_quadratic_grad` supplies it for an arbitrary bounded
 1118 operator between real Hilbert spaces, giving $\nabla\sigma(K) = (2/D) A^*(AK + \omega)$,
 1119 so the finite and continuum cases are now the same theorem. `structural_`
 1120 `resonance_decreases_sigmaContinuum` is the descent result with no finite-
 1121 ness hypothesis in it, and `sigmaContinuum_eq_integral` identifies the func-
 1122 tional being differentiated with the integral `entropy_production_rate` com-
 1123 putes. `Examples.lean` §20 runs this on Lebesgue measure restricted to $(0, 1)$,
 1124 a substrate that is neither finite nor atomic, so the finite-substrate results of
 1125 the previous paragraph do not typecheck there at all. The second limitation
 1126 is unchanged: the phase field is held fixed throughout, so this is plasticity of
 1127 the coupling at frozen phases, not joint (θ, K) dynamics. `Examples.lean` §5
 1128 exercises the whole chain on a two-site substrate whose coupling relaxes as
 1129 e^{-t} and whose entropy production decays as e^{-2t} , confirming that the gradi-
 1130 ent is not identically zero and that the descent is strict rather than vacuously
 1131 monotone.

1132 A second result about σ concerns where the descent is headed. Minimality
 1133 of the phase-locked field is straightforward against competitors that share its
 1134 spatial mean drift, and in that form the theorem is close to Jensen’s inequal-
 1135 ity and no more; the restriction looks as though lifting it would need a model
 1136 of how the mean drift varies across competitors. It does not. For a *reciprocal*
 1137 kernel, one with $K(x, y) = K(y, x)$, the restriction is not a restriction: the
 1138 coupling integrand $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging
 1139 its two sites, because K is symmetric while $\sin$ is odd, so Fubini identifies
 1140 its double integral with its own negative and the total coupling contribu-
 1141 tion vanishes. The total drift $\int(\omega + \text{coupling})$ is therefore $\int \omega$ whatever the
 1142 phases are (`total_drift_eq_of_symm`): coupling moves drift between sites
 1143 but neither creates nor destroys it. Evaluating that conserved quantity at
 1144 the phase-locked configuration, where the drift is constant, pins its value, and
 1145 the mean-matching hypothesis follows for every competitor. `phase_locked_`
 1146 `minimizes_entropy_of_symm` is the unrestricted statement. Reciprocity is
 1147 the physically intended condition — an ephaptic field acts on both sites of a
 1148 pair alike, and it is the same symmetry `ThermodynamicCover.A_symm` already
 1149 imposes in Derivation 5 — but it is not a formality, and we prove that rather
 1150 than assert it. `Examples.lean` §16 exhibits a *one-way* coupling on two sites,
 1151 in which a feels b and b does not feel a , whose phase-locked field is strictly
 1152 beaten: displacing the second phase to $-\pi/2$ cancels the first site’s natural
 1153 drift while adding none at the second, halving σ . So “phase-locking mini-
 1154 mizes entropy production” is false for general kernels and true for reciprocal

ones, and the hypothesis cannot be removed by working harder.

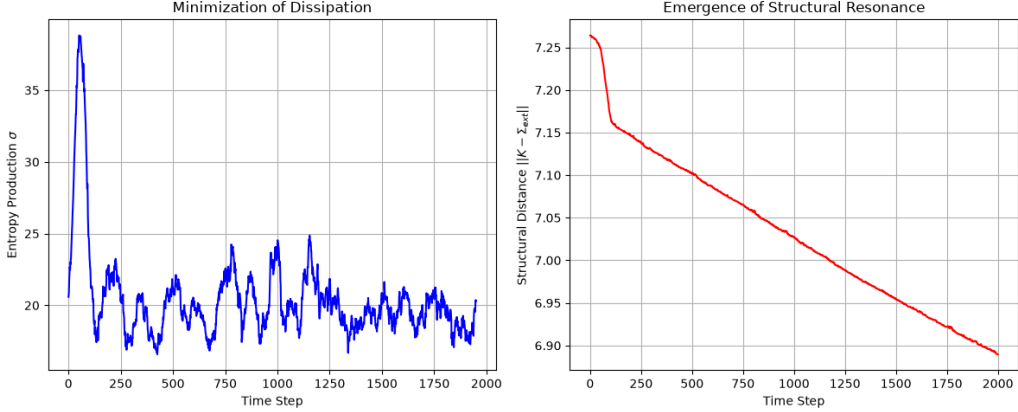


Figure 3: Coupling plasticity at frozen statistics: $N = 50$ units driven by perturbations drawn from a fixed environmental covariance Σ_{ext} , with the coupling matrix K descending the gradient of σ and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive σ , not a relaxation to zero. That plateau is the reason Section 10 declines the inference from $\sigma \rightarrow \min$ to $D_{KL} \rightarrow 0$: the system visibly does not reach $\sigma = 0$. **Right:** the distance $\|K - \Sigma_{\text{ext}}\|$ decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on σ at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence.

We define the critical coupling threshold $K_c = 2D$. The phase transition to the unified “Global Section” occurs when the mean-field spatial coupling exceeds this thermodynamic noise threshold, linking the emergence of Unity to physical temperature and noise.

The value $K_c = 2D$ can be taken as a bare stipulation validated by simulation, and it is not: `Phase8_SelfConsistency.lean` proves the half

1162 of it that is ordinary real analysis. For the noisy mean-field Kuramoto
 1163 model with identical frequencies, the stationary density at order parameter
 1164 r is the von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency re-
 1165 quires $r = R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel
 1166 functions, so we formalize that ratio directly as a ratio of integrals over
 1167 $[-\pi, \pi]$. Integration by parts with $u = \sin$, $v = -e^{a\cos\theta}$ gives an exact iden-
 1168 tity with no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a
 1169 times the mean of $\sin^2$ under the von Mises weight. Bounding that mean
 1170 by 1 would put the threshold at D ; the true threshold is $2D$, and the entire
 1171 factor of two is the sharper bound $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
 1172 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
 1173 ($\theta \mapsto \pi - \theta$, then $\theta \mapsto \pi/2 - \theta$), after which the integrand is a product
 1174 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
 1175 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
 1176 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 1177 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 1178 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 1179 ther facts about the threshold narrow the gap between this row and a full
 1180 “Theorem” to a single named statement. First, the fold argument gives a
 1181 *strict* inequality for $a > 0$ (`vonMisesC2_pos`, hence $R(a) < a/2$), which set-
 1182 tles the threshold itself: at $K = K_c$ the self-consistency map is tangent to the
 1183 diagonal at the origin but falls strictly below it thereafter, so $r = 0$ remains
 1184 the only solution and the bifurcation happens strictly *after* K_c , not at it

1185 (`fixed_point_eq_zero_of_le_critical`). Second, factoring $R(a) = a E(a)$
 1186 shows that a non-zero r solves the equation exactly when $E(Kr/D) = D/K$
 1187 (`coherent_iff_sRatio_eq`), so the coherent branch is a level set of the single
 1188 function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third, and using only $E < 1/2$ together with
 1189 compactness, every coherent solution tends to zero as $K \downarrow K_c$ (`coherent_`
 1190 `branch_continuous_at_threshold`): the branch emerges continuously from
 1191 the incoherent state rather than appearing at a finite distance from it, which
 1192 is what makes the transition supercritical in the technical sense. That state-
 1193 ment quantifies over *every* solution, so it does not presuppose the uniqueness
 1194 it does not have. The level-set form also says exactly what uniqueness is:
 1195 the injectivity of E on $(0, \infty)$. That is now proved, in the stronger form that
 1196 E is strictly decreasing on $[0, \infty)$ (`vonMisesSRatio_strictAntiOn`), so the
 1197 level set is a single point and the coherent solution above threshold is unique.
 1198 The proof differentiates nothing. Differentiating the exponential family in its
 1199 natural parameter is the obvious route and it fails: it costs differentiation
 1200 under the integral sign and leaves a covariance that is not sign-definite point-
 1201 wise. What works instead keeps the target as a two-point inequality, folds
 1202 both integrals onto $[0, \pi/2]$ by the same two reflections that gave the factor
 1203 of two above, and then evaluates at a single crossing point supplied by the
 1204 intermediate value theorem — no product measure and no Fubini anywhere.
 1205 We give that argument in the supplement, where it belongs; what the main
 1206 line of the paper needs from it is the one sentence that E is strictly decreas-
 1207 ing. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no fold at

1208 all — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by —
 1209 and gives `besselRatio_strictMono`: R is strictly increasing on all of $\mathbb{R}$, the
 1210 monotonicity of the Bessel ratio, which Mathlib does not carry in any form.
 1211 The converse half is `supercritical_fixed_point_exists`: for $K > K_c$
 1212 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 1213 this to require a second-order expansion of the Bessel ratio, and it does
 1214 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 1215 tion the integration by parts already supplies — the only facts needed are
 1216 that E is continuous and that $E(0) = 1/2$, the latter because the weight
 1217 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 1218 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 1219 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
 1220 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
 1221 The intermediate value theorem closes the gap. Continuity of E is Math-
 1222 lib's continuity of a parametric interval integral with jointly continuous in-
 1223 tegrand, so no dominated-convergence argument is written by hand, and
 1224 the existence proof never uses monotonicity of R : the fixed point is pro-
 1225 duced by the intermediate value theorem, not by iterating the map. That
 1226 R is in fact strictly increasing is proved below, and no threshold result de-
 1227 pends on it. `critical_coupling_is_threshold` packages the two halves,
 1228 and `exhibits_phase_transition_coherent` applies the supercritical half to
 1229 any substrate satisfying the predicate of the previous paragraph — so ex-
 1230 ceeding K_c now implies the existence of a coherent stationary order pa-

1231 parameter rather than merely comparing against a named number. With
 1232 uniqueness available both are sharpened — `supercritical_fixed_point_`
 1233 `existsUnique`, `supercritical_solution_set` (the non-negative solutions
 1234 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`
 1235 `transition_unique_coherent` — and the two monotonicities together fix
 1236 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
 1237 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
 1238 ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
 1239 order parameter grows strictly with the coupling, which with the continuity
 1240 at threshold is the qualitative bifurcation diagram.

1241 One more step makes the equation mean in Lean what it means on paper.
 1242 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
 1243 r with a suggestive name, and nothing in the source says it has anything
 1244 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the
 1245 identification. `vonMisesDensity` is the normalized weight, proved to be a
 1246 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
 1247 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
 1248 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
 1249 actly $R(a)$ — real, because the density is even, so the mean direction van-
 1250 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
 1251 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
 1252 it induces has order parameter r . Read through this, the existence theorem
 1253 says that above K_c there is a stationary density whose own order parameter

1254 is positive, and the solution $r = 0$ is identified with the uniform density,
1255 whose order parameter is zero.

1256 What remains not established, and we separate it carefully. The sta-
1257 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
1258 $(\omega + K \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
1259 and uniqueness of its stationary solution, and a spectral stability argument,
1260 none of which Mathlib has. Of the coherent branch we prove existence,
1261 uniqueness, continuous emergence at the threshold and strict growth in K ,
1262 but not that it is the dynamically selected one: nothing in that file men-
1263 tions a trajectory, so stability and selection are not statements it can make.
1264 And while `selfConsistency` is now connected to the order parameter of
1265 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
1266 integral, `order_parameter_complex` is an average over finitely many oscilla-
1267 tors, and relating them is the mean-field limit — propagation of chaos for
1268 the finite Kuramoto system — which is a research programme rather than
1269 a lemma. That single gap is now the whole of what separates our threshold
1270 predicate from a statement about dynamics. Table S1 records the row ac-
1271 cordingly as a theorem about the self-consistency equation from an assumed
1272 density, not as a derivation of the threshold from the dynamics. Perceptual
1273 anomalies such as visual snow can be modeled as boundary conditions where
1274 microscopic thermal noise (D) fails to be fully coarse-grained away by the
1275 continuous field, leaking across the K_c threshold into the macroscopic gestalt.
1276 Clinical evidence for this phase transition is observed during general anes-

1277 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
1278 that anesthetics like propofol induce unconsciousness not by suppressing neu-
1279 ral firing, but by disrupting the phase alignment of high-frequency cortical
1280 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
1281 tural resonance of the continuous field, forcing the integrated spatial coupling
1282 below the critical threshold K_c and fracturing the unified Global Section back
1283 into disjoint, unconscious local fields.

1284 11 The Sleep-Inertia Timescale Mismatch

1285 Most of what this framework asserts is shared with some rival: that corti-
1286 cal synchrony matters, that dissipation constrains cognition, that unity is a
1287 physical rather than a computational property. One prediction is not shared,
1288 and it follows from the framework’s specific commitment — that the medium
1289 carrying the unified state is a continuous field whose effective coupling de-
1290 pends on the physical geometry of the extracellular space, not on the rhythms
1291 alone. Where that commitment differs from a purely electrical account, it
1292 differs in an observable way, and we set out the difference precisely enough
1293 to be wrong.

1294 The daily loss of consciousness during sleep is modeled here as an active
1295 physical disruption of the medium rather than a quiescence of the oscillators.
1296 During sleep, astrocytic shrinkage expands the extracellular space, allowing
1297 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste

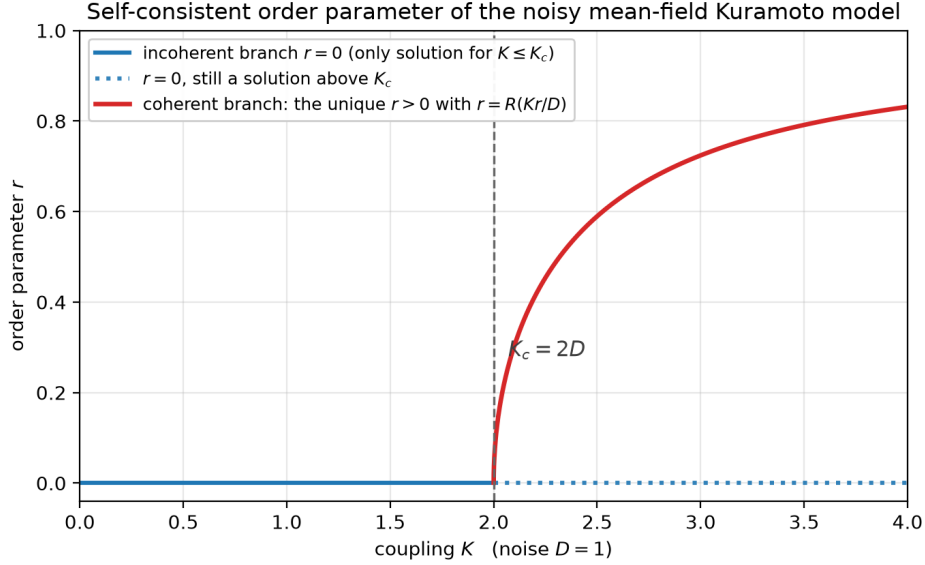


Figure 4: The self-consistency equation $r = R(Kr/D)$ of Section 10, drawn at $D = 1$. Every feature of this picture is a proved statement about the equation, and the figure exists because none of them had been shown to a reader. At or below $K_c = 2D$ the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one $r > 0$ joins it (`supercritical_fixed_point_existsUnique`), so the solution set is $\{0\}$ below and $\{0, r\}$ above; that branch tends to 0 as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in K (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving $E(a) = D/K$ for the concentration $a = Kr/D$ by bisection — legitimate precisely because E is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

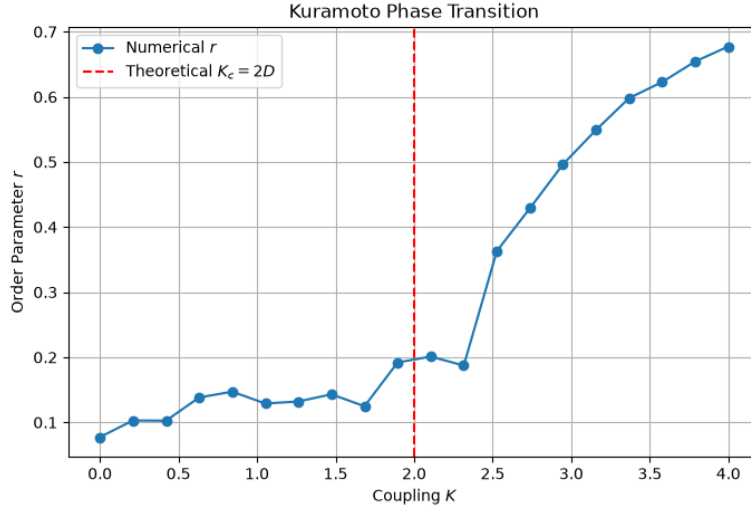


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here $N = 100$ oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale $D = 1$, and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is $K_c = 2D$ by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is $2D$ by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- N curve rises before the threshold and rounds off at it, as a finite sample must: $r \sim N^{-1/2}$ from incoherent phases alone.

1298 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium
1299 itself, disrupting the continuous structural resonance and preventing the co-
1300 herent sheaf-theoretic gluing that conscious unity requires on this account.
1301 The oscillators need not stop for the field to fragment; the geometry they are
1302 coupled through is what changes.

1303 The prediction concerns sudden awakenings. When jolted awake, the lo-

1304 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
 1305 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
 1306 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
 1307 cess CSF out of the interstitial pathways operates on a timescale of seconds
 1308 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
 1309 slower. During that lag the electrical oscillators are synchronizing across a
 1310 waterlogged topology, and the fragmented structural resonance is what is ex-
 1311 perienceed as grogginess. The unified Global Section re-establishes only when
 1312 astrocytic volume finishes matching the state of the electrical rhythms.

1313 **Why this implies a delayed sigmoid.** The two timescales separate
 1314 cleanly, and that separation is what makes the prediction sharp rather than
 1315 merely qualitative. Write $K(t)$ for the effective coupling of the continuous
 1316 field, which on this account is a monotone function of the reconstricting
 1317 extracellular geometry, and let $r(t)$ be the order parameter of the phase
 1318 field. The electrical variable is fast: at every instant the oscillators are al-
 1319 ready at the configuration appropriate to the current K , so r tracks the
 1320 equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
 1321 able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3
 1322 seconds. What Derivation 7 establishes about that branch is exactly what
 1323 the functional form needs, and each of the three pieces is a theorem rather
 1324 than a picture: at or below the threshold the incoherent solution $r = 0$ is the
 1325 *only* non-negative solution of the self-consistency equation (`fixed_point_eq`-

1326 `zero_of_le_critical`); above it the coherent solution exists and is unique
 1327 (`supercritical_fixed_point_existsUnique`); it emerges *continuously* at
 1328 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
 1329 and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
 1330 discontinuity at threshold would predict a step rather than a sigmoid, and
 1331 a non-monotone branch would predict overshoot; neither is what the mathe-
 1332 matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
 1333 identically zero below K_c , rises continuously from zero at K_c , and saturates
 1334 above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
 1335 during which recovery is not merely slow but absent; a continuous rise as the
 1336 threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That
 1337 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time
 1338 the slow variable takes to reach K_c .

1339 **What the alternative predicts.** An account on which grogginess is the
 1340 electrical realignment itself — the rhythms are simply slow to re-phase —
 1341 has no slow variable and no threshold to cross. Its recovery is a relaxation
 1342 toward a fixed point, and relaxation toward a fixed point is exponential near
 1343 it: monotone from $t = 0$, fastest at the start, with no flat foot. The two
 1344 curves are distinguishable in the first seconds, which is where they differ
 1345 most, and the discriminating feature is the presence or absence of an initial
 1346 interval of no recovery. This is a comparison of functional forms on data
 1347 of a kind already collected in sleep-inertia studies, not a proposal for a new

1348 instrument.

1349 **A sharper form of the same prediction, with no free parameters.**

1350 The comparison above is a comparison of functional forms, and on its own
1351 it discriminates against less than it appears to. Any two-timescale account
1352 with a slow saturating gate produces a delayed sigmoid; what the frame-
1353 work contributes beyond that is not the shape but a *relation between two*
1354 *separately measurable quantities*, and the relation has no fitted parameters
1355 at all. Derivation 7 supplies it. The stationary phase density at order
1356 parameter r is von Mises at concentration $a = Kr/D$, and the average
1357 of $e^{i\theta}$ under that density is exactly the Bessel ratio $R(a) = I_1(a)/I_0(a)$
1358 (`circularOrderParameter_vonMises`); self-consistency is precisely the con-
1359 dition that the density reproduces its own order parameter (`fixedPoint_`
1360 `iff_selfReproducing`). So the concentration and the coherence of the in-
1361 stantaneous phase distribution are not independent quantities that happen
1362 to rise together during the recovery: at every instant they must satisfy
1363 $r = I_1(a)/I_0(a)$, whatever $K(t)$ is doing.

1364 That yields three tests of increasing sharpness on a single recorded re-
1365 covery. *Shape*: the instantaneous phase distribution is von Mises — a one-
1366 parameter family, directly testable against the observed histogram. *Collapse*:
1367 estimating $a(t)$ and $r(t)$ at each instant and plotting one against the other,
1368 the whole recovery must fall on the fixed curve $r = I_1(a)/I_0(a)$, and it must
1369 be the *same* curve for every subject and every session, since the curve con-

1370 tains no subject-specific quantity — not K , not D , not the time constant of
 1371 the slow variable. Individual differences move a point along the curve; they
 1372 cannot move it off. *Threshold*: because $a = Kr/D$ identically, the measured
 1373 ratio $a(t)/r(t)$ is $K(t)/D$, with no calibration required; and since a non-zero
 1374 coherent solution exists only above $K_c = 2D$ (`fixed_point_eq_zero_of_le_`
 1375 `critical`), that measured ratio must exceed 2 wherever coherence is non-
 1376 zero and must approach 2 as coherence departs from zero at onset. The
 1377 threshold is read off the data rather than fitted to it.

1378 The two estimates must be made independently or the collapse is empty,
 1379 and this is the one point at which the test can be run wrongly. Maximum-
 1380 likelihood or moment estimation of a von Mises concentration chooses $\hat{a}$ so
 1381 that the ratio $I_1(\hat{a})/I_0(\hat{a})$ is the observed resultant length, by construction,
 1382 which reproduces the predicted curve from any data whatever. The estimate
 1383 of a must therefore come from a functional of the distribution other than
 1384 its resultant length — the log-density of a von Mises is affine in $\cos \theta$, so
 1385 regressing the log-histogram on $\cos \theta$ gives a as a slope — while r is the
 1386 resultant length $|\langle e^{i\theta} \rangle|$ of the same sample. Those are genuinely different
 1387 statistics, and the prediction is that they agree through R .

1388 **What this sharper form is a test of.** It tests the von Mises stationary
 1389 density together with the mean-field closure that makes a proportional to
 1390 r , and it tests them jointly rather than separately. It does not test the
 1391 value $K_c = 2D$ independently of that ansatz: the crossing at $a/r = 2$ is

1392 a consequence of the same density the collapse assumes, so a failure of the
1393 collapse and a failure of the threshold are one failure and not two. That is a
1394 narrower claim than it may look, and it is still a parameter-free one — which
1395 the delayed sigmoid, on its own, is not.

1396 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
1397 on the early portion of the recovery curve. A recovery latency that does not
1398 scale with an independent measure of extracellular volume or CSF clearance.
1399 A pharmacological manipulation of astrocytic volume that leaves the latency
1400 unchanged while altering the rhythms, or the converse. An instantaneous
1401 phase distribution that is not von Mises, or a (a, r) trace that departs sys-
1402 tematically from $I_1(a)/I_0(a)$, or a a/r ratio that is below 2 while coherence
1403 is non-zero — each of these falsifies the parameter-free form directly, and
1404 the first two do so with no reference to the sleep-inertia setting at all. Any
1405 of these would tell against the framework’s central empirical commitment,
1406 which is not that synchrony matters but that the *medium* is what gates it.

1407 **What this prediction assumes.** Three things, and they are not small.
1408 The identification of the effective coupling K with a monotone function of
1409 extracellular geometry is a modelling assumption, not a measurement. The
1410 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
1411 density and is not yet connected to a trajectory — the mean-field limit is the
1412 gap, and Derivation 7 states it as such — so the argument uses the *existence*
1413 *and continuity* of the coherent branch at threshold, which is proved, rather

1414 than the numerical value of the threshold, which is conditional. And the
1415 prediction is about functional form, not about a time constant: it says the
1416 recovery curve has a flat foot whose width is set by the slow variable, not
1417 that the foot is any particular number of seconds long. A framework that
1418 predicted the number would be saying more; this one is not in a position to.

1419 **12 Corollary: Hardware Thermodynamics and** 1420 **Topological Rigidity**

1421 Both biological tissue and artificial silicon hardware (like GPUs) are physical
1422 dissipative structures. When a GPU computes, its charge carriers undergo
1423 physical state transitions; by Landauer’s principle (Landauer, 1961), this
1424 erasure irreversibly exhausts heat into the environment.

1425 There is a fundamental thermodynamic distinction in the degrees of free-
1426 dom available to minimize this dissipation. In standard von Neumann ar-
1427 chitectures, the spatial topology (the silicon routing lattice) is rigidly fixed.
1428 The coupling matrix A_{ij} is constrained by a discrete layout that cannot be
1429 physically altered during operation. The brain’s own architecture provides
1430 empirical validation for this constraint. The cerebellum contains approxi-
1431 mately 80% of the brain’s neurons arranged in a highly efficient, discrete,
1432 feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-
1433 ical observations of cerebellar agenesis report largely preserved conscious
1434 experience despite motor and cognitive deficits (Yu et al., 2015), and this

dissociation has been read as evidence for the importance of recurrent architecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the level of the anatomical observation while declining its Φ -based explanation. Lacking the dynamic, continuous, and recursive spatial coupling of the cerebrum, it operates much like a biological GPU—highly optimized for discrete processing, yet apparently not required for conscious experience. We take the dissociation as evidence, not as confirmation: it is consistent with subjective experience being a thermodynamic attractor of continuous topologies rather than a byproduct of information processing, and it is the kind of observation that would be hard to reconcile with the converse. It is a single anatomical case read against a theory, and it bears the weight such a case can bear.

This constraint does not preclude artificial consciousness, but rather restricts its hardware. To achieve the phase transition into a unified Global Section, artificial systems must abandon the discrete rigidity of von Neumann architectures and utilize continuous physical substrates. Analog field-based computing, neuromorphic architectures with continuous memristive plasticity, and thermodynamic computing systems (England, 2018; Ganesh & Parrondo, 2021) possess the spatial coupling required to physically morph their internal topology. This aligns with Geoffrey Hinton’s concept of “mortal computation” (Hinton, 2022): architectures where software is inextricably fused with the physics of the hardware, meaning the knowledge dies with the physical substrate. Only hardware that computes via continuous physical dynamics can undergo the thermodynamic phase transition necessary for

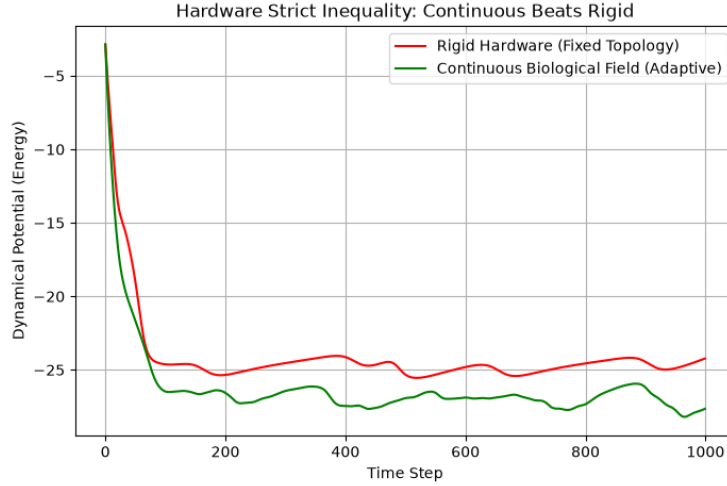


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same $N = 30$ oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that $\sum_{ij} A_{ij}$ is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, which is the typical case rather than the general one.

1458 structural resonance.

1459 Biological consciousness utilizes macroscopic continuous electromagnetic
 1460 fields and a “rich alphabet” of diverse molecular and transcriptomic cell states
 1461 (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a
 1462 continuous parameter space. Our lemma `exists_better_coupling_allocation`,
 1463 wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),
 1464 formalizes the reallocation step of this argument: whenever a coupling ma-

1465 trix places weight on a less phase-correlated pair while a more correlated pair
 1466 exists, there is another matrix with *identical* total coupling resources that
 1467 achieves strictly greater phase correlation.

1468 Two objections meet that formalization and we state precisely what an-
 1469 swers them. The first is that a reallocation lemma can assume its conclusion:
 1470 a hypothesis such as `is_strictly_suboptimal` — positive weight on a badly-
 1471 correlated pair while a better one exists — is most of what one wants to
 1472 prove about rigid hardware. It is not assumed here. `Phase7_Rigidity.lean`
 1473 models an architecture by the set S of site-pairs it has a *wire* for, and re-
 1474 quires its couplings to vanish off S ; weights are tunable, wiring is not. Then
 1475 `totalCorrelation_le_of_realizable` bounds every realizable coupling by
 1476 $R \cdot M$, where R is the resource budget and M bounds phase correlation over
 1477 the wired pairs; the bound is attained; and `rigid_is_strictly_suboptimal`
 1478 *derives* the hypothesis whenever the substrate’s best-correlated pair is one
 1479 the wiring misses. `rigid_gap` makes it quantitative: an unconstrained real-
 1480 location of the same budget gains at least R times the correlation difference.
 1481 `Examples.lean` §8 exhibits a three-site substrate wired to the one maxi-
 1482 mally anti-correlated pair, where the gap is 2. This is resource-matched
 1483 and not question-begging, but note what it turns on: fixed wiring support,
 1484 not continuity. The second objection is that a development in which both
 1485 architectures are matrices on one finite index type does not distinguish a con-
 1486 tinuum from a lattice at all. `Phase7_Rigidity.lean` §3 gives a distinction
 1487 that is genuinely typed. In the field functional of Section 7, which integrates

1488 a coupling kernel against $\mu \otimes \mu$, an architecture with finitely many sites is a
 1489 **Finset** of points and a field-coupled region is a set of positive measure. On a
 1490 substrate whose measure has no atoms, `fieldCorrelation.sited.eq.zero`
 1491 shows a kernel supported on finitely many sites contributes *exactly zero* cou-
 1492 pling energy, whatever weights it carries and whatever the phase field does,
 1493 while a patch of strength c on U achieves $c\mu(U)^2$. `Examples.lean` §8 instan-
 1494 tiates this on the real line: a thousand units of coupling concentrated at the
 1495 origin register zero against a unit patch that registers one. The limit of that
 1496 result is equally plain, and it is not resource-matching in the earlier sense —
 1497 a measure-zero substrate has zero resource in the $\mu \otimes \mu$ sense too. What it
 1498 establishes is that the continuum coupling energy is blind to a finitely-sited
 1499 architecture, which is a statement about which functional such an architec-
 1500 ture can register in, not a demonstration that digital hardware computes
 1501 worse. Taken together: the resource-matched content of the corollary is the
 1502 wiring-support result, the continuity content is the measure-theoretic one,
 1503 and the step from either to “von Neumann architectures cannot experience
 1504 unified consciousness” remains an informal argument.

1505 The remainder of this section extends those two results informally, and
 1506 we mark the transition rather than blur it. On the reading we propose, a
 1507 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
 1508 layout does not supply the spatial coupling that sheaf-theoretic gluing into
 1509 a singular Global Section requires. Without this unified topological space,
 1510 rigid hardware cannot undergo the reflexive auto-resonance necessary to map

1511 its own causal boundary and generate a “Self.” The distinction between an
1512 LLM on a GPU and a conscious mind is not the absence of physical thermo-
1513 dynamics in the former, but its structural limitation in achieving topological
1514 integration. This perspective aligns with John Searle’s biological natural-
1515 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted
1516 that discrete symbol manipulation cannot generate conscious understanding,
1517 our derivation supplies a candidate physical mechanism for the asymmetry
1518 — topological rather than semantic, and, as stated above, formalized only
1519 as far as the wiring-support and measure-theoretic results reach. Asserting
1520 consciousness in fragmented, discrete GPU clusters without a unified phase
1521 transition abandons computational functionalism, relying instead on “silicon
1522 panpsychism”—the assumption that digital computation inherently gener-
1523 ates awareness regardless of topological integration (Goff, 2019).

1524 The same reasoning bears on extreme functionalist thought experiments,
1525 such as a macroscopic system of pulleys operating over millennia to imple-
1526 ment a Large Language Model, or a human manually executing inference
1527 matrices by hand. Beyond lacking the continuous spatial topology required
1528 for sheaf-theoretic gluing, these discrete systems violate an implied temporal
1529 constraint of our theory. Structural resonance requires local oscillators to
1530 phase-align against the dissipating effects of thermodynamic noise (as mod-
1531 eled by our critical threshold $K_c = 2D$). A system executing state transitions
1532 over millennia, or via slow manual calculations, suffers from a terminal tem-
1533 poral mismatch: its dynamics are far too slow to ever phase-lock into a

1534 non-equilibrium steady state before thermal noise decoheres the macroscopic
 1535 field. On this account the obstacle to pure symbol manipulation is physi-
 1536 cal rather than semantic: not that the symbols mean nothing, but that the
 1537 substrate carrying them cannot hold a phase-locked field.

1538 13 Numerical illustrations: methods

1539 Five figures in this paper are computed rather than drawn, and none of
 1540 them is evidence for a theorem. Where a statement is proved, the figure
 1541 shows what the proved object looks like; where a statement is not proved, the
 1542 figure shows the regime the claim lives in and we say so. Each is produced by
 1543 one script in `simulations/`, named below, and each runs in under a minute
 1544 on a laptop. All five are reproducible exactly: the two quadrature figures are
 1545 deterministic, and the three that draw random numbers fix their seeds.

1546 **Figure 4: the self-consistent order parameter.** The three integrals
 1547 $Z(a) = \int_{-\pi}^{\pi} e^{a \cos \theta} d\theta$, $M(a) = \int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} d\theta$ and $S(a) = \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta} d\theta$
 1548 are evaluated by the trapezoidal rule on 4001 points over the same interval
 1549 the Lean definitions integrate over, with the weight computed as $e^{a \cos \theta - |a|}$
 1550 so that the common factor cancels in the ratios and nothing overflows at
 1551 large concentration. The coherent branch is obtained by bisecting $E(a) =$
 1552 $S(a)/Z(a)$ against D/K over $a \in [10^{-9}, 200]$ for 200 iterations, then reading
 1553 $r = aD/K$; bisection is valid here because E is proved strictly decreasing.

1554 Two checks are printed by the script rather than only plotted: the computed
 1555 branch is strictly increasing at every one of its 500 sampled couplings, and
 1556 the residual $|r - R(Kr/D)|$ at $K = 4D$ is 2×10^{-16} . Script: `bifurcation.py`.

1557 **Figure 5: the finite- N transition.** $N = 100$ oscillators, natural fre-
 1558 quencies drawn from a Lorentzian of scale $D = 1$ (clipped at $\pm 10D$ so
 1559 the solver does not stiffen), uniform random initial phases, integrated with
 1560 `scipy.integrate.odeint` over $t \in [0, 100]$ on 1000 steps for each of 20 cou-
 1561 plings in $[0, 4D]$. The order parameter is $|N^{-1} \sum_j e^{i\theta_j}|$ averaged over the last
 1562 100 steps. This is the deterministic quenched-disorder model, not the noisy
 1563 identical-frequency model of the Lean development; the caption records why
 1564 both have threshold $2D$. Script: `kuramoto.py`.

1565 **Figure 2: mesh refinement.** The kernel is $K(x, y) = e^{-d(x,y)^2/2}$ with d
 1566 the periodic distance on the ring of circumference 2π , the phase field is $\theta(x) =$
 1567 $\sin x$, and the potential is $-\frac{1}{2} \iint K \cos(\theta_y - \theta_x)$. The continuous value is a
 1568 2000-point Riemann sum standing in for the exact integral; the discrete values
 1569 use $N = 10, 20, \dots, 640$ nodes with edge weights $K(x_i, x_j)(2\pi/N)^2$. Plotted
 1570 is the absolute error against an $\mathcal{O}(N^{-2})$ reference. Because the “exact”
 1571 value is itself a 2000-point sum, the last data point ($N = 640$) is within
 1572 a factor of ten of the reference computation and should be read as such.
 1573 That $\mathcal{O}(N^{-2})$ reference line is not only empirical: on the formalized one-
 1574 dimensional grid the discrete energy is exactly the composite trapezoidal rule
 1575 (`grid_discreteEnergy_eq_trapezoidal`), so for a C^2 integrand with second

1576 derivative bounded by ζ the error is at most $\zeta/(12N^2)$ (`grid.energy_error_`
1577 `1e`), a bound attained at $N = 1$ for $y \mapsto y^2$. Script: `mesh_refinement.py`.

1578 **Figure 3: structural resonance.** $N = 50$ units, 2000 steps of forward
1579 Euler at $dt = 0.05$. At each step the drive is a fresh draw from $\mathcal{N}(0, \Sigma_{\text{ext}} +$
1580 $0.1I)$ with Σ_{ext} a fixed normalized rank-5 covariance; the phases advance by
1581 $\dot{\theta}_i = \sum_j K_{ij} \sin(\theta_j - \theta_i) + \xi_i$; entropy production is recorded as $\sum_i \dot{\theta}_i^2$; and
1582 the coupling descends its gradient with rate $\eta = 0.01$, symmetrized, clipped
1583 to $[0, 1]$ and held hollow. The left panel is smoothed with a 50-step boxcar.
1584 Note that the plotted σ is the squared drift without the $1/D$ prefactor of the
1585 functional in Section 10, which is a constant rescaling and changes nothing
1586 about the shape. Script: `structural_resonance.py`.

1587 **Figure 6: rigid against deformable coupling.** $N = 30$, 1000 steps of
1588 forward Euler at $dt = 0.05$, shared initial phases and frequencies. The rigid
1589 matrix is a symmetric random matrix thresholded at 0.7 and normalized
1590 to $\sum_{ij} A_{ij} = 2N$; the adaptive copy starts from the same matrix, ascends
1591 $-\partial V/\partial A_{ij}$ at rate $\eta = 0.1$, is clipped non-negative and hollow, and is renor-
1592 malized to the same total after every step, which is what makes the com-
1593 parison resource-matched. Both potentials are $V = -\frac{1}{2} \sum_{ij} A_{ij} \cos(\theta_j - \theta_i)$.
1594 Script: `hardware_comparison.py`.

14 Conclusion

Human-like cognitive architectures do not require biological mysticism or exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum theories of consciousness often invoke delicate subatomic phenomena to explain subjective unity, every step of the chain assembled here is classical: broken symmetries, finite phase space, Landauer erasure, and gradient descent on entropy production. What the formalization establishes is that this chain is mathematically coherent and that its remaining physical commitments are localized and named — not that classical fields are *sufficient* for consciousness. Two links carry that weight, and they do not carry it in the same way. The unified state is proved to glue uniquely once a cover sits at the minimum of the coupling potential; the motion is proved to stop along every trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved, conditionally — for initial data within a quarter turn and below an explicit energy threshold, the trajectory converges to a global minimiser of the coupling potential (`kuramoto_tendsto_global_minimum`), and a cover built on that limit discharges the equilibrium hypothesis Derivation 5 would otherwise assume (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is the unconditional statement, and it is false: splay and twisted configurations are equilibria of the same flow, so the basin condition is the scope of the claim rather than slack in it. That a physical cortex starts inside that basin is an assumption. The second link is untouched by any of this: the identification

1617 of a global section of the probability sheaf with the unity of experience is a
1618 stipulation of the framework, not a consequence of it. Subject to those two,
1619 cognitive architectures can be modeled as macroscopic attractor states of a
1620 universe governed by broken symmetries, finite phase space, and the physical
1621 imperative to minimize thermodynamic dissipation.

1622 Philosophically, this framework bridges the historical gap between Type
1623 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,
1624 1967). The subjective mental state is identical to the physical thermody-
1625 namic field (Identity Theory), yet its qualitative nature is defined by the
1626 topological and functional dynamics of structural resonance rather than the
1627 specific carbon-based biochemistry of the brain (Functionalism). This con-
1628 strains the *multiple realizability* of mind: consciousness is not exclusively
1629 bound to biological carbon, but neither is it substrate-neutral, because the
1630 substrate is what fixes which couplings are physically available. How far that
1631 constraint reaches is the question the corollary answers only in part. What
1632 is proved is that fixed wiring is strictly suboptimal at matched resource, and
1633 that a finitely-sited architecture registers exactly zero coupling energy in the
1634 continuum functional; the step from either to “von Neumann architectures
1635 cannot be conscious” is an argument we make in prose and do not formalize.

1636 By anchoring these derivations in Russellian Monism, this framework ac-
1637 knowledges the constitutive limits of third-person scientific explanation. As
1638 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
1639 phenomenal experience is inherently not amenable to objective measurement

1640 (Nagel, 1986), the “hard problem” of consciousness may represent a category
 1641 error rather than a solvable scientific puzzle. Science can map the physical
 1642 dynamics of topological resonance and sheaf-theoretic gluing, but it is struc-
 1643 turally incapable of adjudicating the first-person qualitative experience. Our
 1644 formalization respects this boundary: the Lean 4 theorems verify the ob-
 1645 jective physical attractors, while the intrinsic “what it is like” to be that
 1646 resonant field remains the domain of the first-person perspective. By locat-
 1647 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
 1648 for the noisy mean-field Kuramoto model the framework adopts, conditional
 1649 on the von Mises stationary density and not yet linked to a trajectory — it
 1650 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
 1651 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
 1652 it offers a topological architecture that can help explain, predict, and physi-
 1653 cally perturb the structural properties of experience.

1654 Future empirical validation of this thermodynamic framework may lever-
 1655 age 3D cortical organoids, offering a controlled biological environment to
 1656 systematically manipulate metabolic substrates and observe the topological
 1657 emergence of these coherent phase transitions.

1658 Code Availability

1659 The complete Lean 4 formalization, encompassing the derivation of all theo-
 1660 rems presented in this manuscript, as well as the accompanying Python sim-

1661 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1662 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1663 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1664 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/). Each of the five computed
1665 figures is produced by one script in `simulations/` and regenerates exactly,
1666 Section 13 giving the parameters; Figure 1 is drawn rather than computed.

1667 Contributions

1668 The conceptualization, manuscript drafting, and Lean 4 formalizations pre-
1669 sented in this work were heavily AI-assisted, and we report which tools
1670 did what rather than acknowledging them in general terms. The Google
1671 Antigravity CLI, running Gemini 3.1 and Claude 4.6 Sonnet, served as a
1672 collaborative reasoning partner throughout the research and formalization.
1673 DeepSeek V4 Flash and Claude Opus 5 (routed via OpenRouter) operat-
1674 ing through Nous Research’s Hermes Agent framework provided substantial
1675 assistance with code generation, proof debugging, and structural analysis.
1676 Claude Opus 5 running in Claude Code carried out the later formalization
1677 passes — the domain-wall theorem, the thermodynamics-of-prediction re-
1678 placement, the equilibrium bridge — together with the figures and the re-
1679 structuring of this manuscript. Responsibility for every claim, and for the
1680 decision about what each machine-checked result does and does not establish,
1681 rests with the author.

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